OECD Health

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Table of contents

1 OECD health status

Source of data : https://data-explorer.oecd.org/

OECD health에 있는 주요 데이터 항목들

- Health Expenditure and financing
- Health status
- Risk factors for health
- Healthcare resources and equipment
- Healthcare use
- Pharmaceutical market
- Healthcare coverage
- Subnational health indicators

이들 항목들 중에서 다룰 항목은 Health Expenditure and financing와 Health status인데 이중에서 Health status에 대해서만 먼저 다루도록 한다.

각 항목별로 데이터를 다운로드 받아서 working directory에 저장한 상태이다.

1.1 Cancer Data

파일리스트 중에서 Cancer 항목만 찾아보자. DF_C 라는 이름이 들어가 있다.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
setwd("G:/R project/OECDhealth")
file_list <- dir()
file_list[regexpr("DF_C,", file_list)>0] -> pathname
oecd_cancer <- read.csv(pathname)
oecd_cancer %>% str
```

```
'data.frame':   1668 obs. of  44 variables:
 $ STRUCTURE                : chr  "DATAFLOW" "DATAFLOW" "DATAFLOW" "DATAFLOW" ...
 $ STRUCTURE_ID             : chr  "OECD.ELS.HD:DSD_HEALTH_STAT@DF_C(1.0)" "OECD.ELS.HD:DSD_HEALTH_STAT@DF
 $ STRUCTURE_NAME           : chr  "Cancer" "Cancer" "Cancer" "Cancer" ...
 $ ACTION                   : chr  "I" "I" "I" "I" ...
 $ REF_AREA                 : chr  "ISL" "ISL" "ISL" "IND" ...
 $ Reference.area           : chr  "Iceland" "Iceland" "Iceland" "India" ...
 $ FREQ                     : chr  "A" "A" "A" "A" ...
 $ Frequency.of.observation : chr  "Annual" "Annual" "Annual" "Annual" ...
 $ MEASURE                  : chr  "CNCR" "CNCR" "CNCR" "CNCR" ...
 $ Measure                  : chr  "Cancer incidence" "Cancer incidence" "Cancer incidence" "Cancer incid
 $ UNIT_MEASURE             : chr  "CS_10P5PS" "CS_10P5PS" "CS_10P5PS" "CS_10P5PS" ...
 $ Unit.of.measure          : chr  "Cases per 100 000 persons" "Cases per 100 000 persons" "Cases per 100
 $ AGE                      : chr  "_T" "_T" "_T" "_T" ...
 $ Age                      : chr  "Total" "Total" "Total" "Total" ...
 $ SEX                      : chr  "F" "F" "F" "F" ...
 $ Sex                      : chr  "Female" "Female" "Female" "Female" ...
 $ SOCIO_ECON_STATUS        : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Socio.economic.status    : chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ DEATH_CAUSE              : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Cause.of.death           : chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ CALC_METHODOLOGY         : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Calculation.methodology  : chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ GESTATION_THRESHOLD      : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Gestation.period.threshold: chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ HEALTH_STATUS            : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Health.status            : chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ DISEASE                  : chr  "_Z" "_Z" "_Z" "_Z" ...
 $ Disease                  : chr  "Not applicable" "Not applicable" "Not applicable" "Not applicable" ..
 $ CANCER_SITE              : chr  "CICDCANC" "CICDCANC" "CICDCANC" "CICDTUME" ...
```

```
 $ Cancer.site              : chr  "Malignant neoplasms of trachea, bronchus, lung" "Malignant neoplasms 
 $ TIME_PERIOD              : int  2002 2008 2012 2012 2002 2008 2012 2002 2008 2012 ...
 $ Time.period              : logi  NA NA NA NA NA NA ...
 $ OBS_VALUE                : num  30 27.8 28.9 97.4 26.5 26 34.7 36.1 36.2 33.7 ...
 $ Observation.value        : logi  NA NA NA NA NA NA ...
 $ DECIMALS                 : logi  NA NA NA NA NA NA ...
 $ Decimals                 : logi  NA NA NA NA NA NA ...
 $ OBS_STATUS               : logi  NA NA NA NA NA NA ...
 $ Observation.status       : logi  NA NA NA NA NA NA ...
 $ OBS_STATUS2              : logi  NA NA NA NA NA NA ...
 $ Observation.status.2     : logi  NA NA NA NA NA NA ...
 $ OBS_STATUS3              : logi  NA NA NA NA NA NA ...
 $ Observation.status.3     : logi  NA NA NA NA NA NA ...
 $ UNIT_MULT                : logi  NA NA NA NA NA NA ...
 $ Unit.multiplier          : logi  NA NA NA NA NA NA ...
```

각각의 변수들을 살펴보면 그 값이 단일한 경우가 대부분이다. unique value가 1보다 큰 항목이 몇개가 있는지 살펴보자 Reference.area, Sex , Cancer.site, TIME_PERIOD, OBS_VALUE 다섯가지만 가지고 집계를 하면 될 것으로 생각된다.

```r
for (i in 1:length(oecd_cancer)) {
  if (length(unique(oecd_cancer[,i]))>1)
    {cat(i,")",length(unique(oecd_cancer[,i]))," ",colnames(oecd_cancer)[i],"\n")}
  }
```

```
5 ) 44   REF_AREA
6 ) 44   Reference.area
15 ) 3   SEX
16 ) 3   Sex
29 ) 6   CANCER_SITE
30 ) 6   Cancer.site
31 ) 5   TIME_PERIOD
33 ) 984   OBS_VALUE
```

위의 작업을 checkvar라는 함수로 집어 넣어보자

```r
checkvar <- function(df=oecd_cancer){
  for (i in 1:length(df)) {
    if(length(unique(df[,i]))>1){
      cat(i,")", length(unique(df[,i]))," ",colnames(df)[i],"\n")
    }
}
}
checkvar(oecd_cancer)
```

```
5 ) 44   REF_AREA
6 ) 44   Reference.area
```

```
15 ) 3   SEX
16 ) 3   Sex
29 ) 6   CANCER_SITE
30 ) 6   Cancer.site
31 ) 5   TIME_PERIOD
33 ) 984   OBS_VALUE
```

unique value가 하나이고 NA code가 아닌 경우의 value를 보여주는 함수도 만들어 보자

```r
checkvar2 <- function(df=oecd_cancer){
  for (i in 1:length(df)) {
    if(length(unique(df[,i]))==1&&
       !(is.na(unique(df[,i]))||unique(df[,i]=="Not applicable")||unique(df[,i]=="_Z")))
      {
      cat(i,")", colnames(df)[i],":",unique(df[,i]),"\n")
    }
  }
}
checkvar2(oecd_cancer)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_C(1.0)
3 ) STRUCTURE_NAME : Cancer
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : CNCR
10 ) Measure : Cancer incidence
11 ) UNIT_MEASURE : CS_10P5PS
12 ) Unit.of.measure : Cases per 100 000 persons
13 ) AGE : _T
14 ) Age : Total
```

value가 하나인 변수들은 현재 데이터베이스의 의미를 알려준다.

위의 내용을 보면 annual cancer incidence 를 10만명당 발생률로 전연령대에서 조사한 값이라는 의미를 알 수 있다.

Reference.area = 국가,

Sex = Male, Female, Total ,

Cancer.site = total and 5 sites

TIME_PERIOD = years

OBS_VALUE : incidence in 100,000

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.0     v readr     2.1.5
```

```
v ggplot2   3.5.1     v stringr   1.5.1
v lubridate 1.9.4     v tibble    3.2.1
v purrr     1.0.4     v tidyr     1.3.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
cancerdf <- oecd_cancer[,c(6,16,30,31,33)]
cancerdf <- cancerdf[cancerdf$Sex=="Total",]
sites <- sort(unique(oecd_cancer$Cancer.site))
cancer_incid <- spread(data=cancerdf[cancerdf$Cancer.site==sites[1],], key=TIME_PERIOD, value = OBS_VALU
cancer_incid <- cancer_incid[,c(1,4:8)]
cancer_incid
```

```
                  Reference.area  1998  2000  2002  2008  2012
1                      Australia    NA    NA 312.0 314.1 323.0
2                        Austria 249.4 250.7 275.5 250.6 254.1
3                        Belgium 277.1 282.2 296.2 309.4 321.1
4                         Brazil    NA    NA    NA    NA 205.5
5                         Canada    NA    NA 299.9 296.6 295.7
6                          Chile    NA    NA    NA 176.7 175.7
7   China (People's Republic of)    NA    NA    NA    NA 174.0
8                       Colombia    NA    NA    NA    NA 160.6
9                     Costa Rica    NA    NA    NA    NA 179.3
10                       Czechia    NA    NA 289.3 288.5 293.8
11                       Denmark 263.7 287.8 281.4 321.1 338.1
12                       Estonia    NA    NA    NA 230.1 242.8
13                       Finland 246.5 241.0 246.0 250.1 256.8
14                        France 269.6 281.5 289.5 300.4 303.5
15                       Germany 238.6 273.8 283.3 282.1 283.8
16                        Greece 195.5 191.5 203.0 162.0 163.0
17                       Hungary    NA    NA 318.8 286.6 285.4
18                       Iceland    NA    NA 286.5 290.4 284.3
19                         India    NA    NA    NA    NA  94.0
20                     Indonesia    NA    NA    NA    NA 133.5
21                       Ireland 251.1 254.6 254.3 317.0 307.9
22                        Israel    NA    NA    NA 288.3 283.2
23                         Italy 247.8 254.1 276.5 274.3 278.6
24                         Japan    NA    NA 214.5 201.1 217.1
25                         Korea    NA    NA 220.5 262.4 307.8
26                        Latvia    NA    NA    NA    NA 246.8
27                     Lithuania    NA    NA    NA    NA 251.9
28                    Luxembourg 257.0 271.2 296.7 284.0 280.3
29                        Mexico    NA    NA 147.3 128.4 131.5
30                   Netherlands 262.4 276.5 283.0 289.9 304.8
31                   New Zealand    NA    NA 331.0 309.2 295.0
32                        Norway    NA    NA 286.0 297.9 318.3
```

```
33                         Poland    NA    NA 256.1 225.1 229.6
34                       Portugal 224.5 231.6 237.9 223.2 246.2
35                         Russia    NA    NA    NA    NA 204.3
36                Slovak Republic    NA    NA 263.0 260.6 276.9
37                       Slovenia    NA    NA    NA 264.8 296.3
38                   South Africa    NA    NA    NA    NA 187.1
39                          Spain 233.9 219.3 243.4 241.4 249.0
40                         Sweden 249.1 256.1 264.7 259.7 270.0
41                    Switzerland    NA 247.3 284.6 269.3 287.0
42                        Türkiye    NA    NA 114.3 144.8 205.1
43                 United Kingdom 240.3 247.3 273.6 269.4 272.9
44                  United States    NA    NA 357.7 300.2 318.0
```

1.2 Health Status

Status 라는 이름이 들어가는 화일을 읽는다.

```r
file_list <- dir()
file_list[regexpr("STATUS", file_list)>0]
```

```
[1] "OECD.ELS.HD,DSD_HEALTH_STAT@DF_HEALTH_STATUS,1.0+all.csv"
```

```r
file_list[regexpr("STATUS", file_list)>0] %>% read.csv -> status
checkvar(status)
```

```
5 ) 49    REF_AREA
6 ) 49    Reference.area
9 ) 19    MEASURE
10 ) 19    Measure
11 ) 13    UNIT_MEASURE
12 ) 13    Unit.of.measure
13 ) 12    AGE
14 ) 12    Age
15 ) 3    SEX
16 ) 3    Sex
17 ) 6    SOCIO_ECON_STATUS
18 ) 6    Socio.economic.status
19 ) 56    DEATH_CAUSE
20 ) 56    Cause.of.death
21 ) 3    CALC_METHODOLOGY
22 ) 3    Calculation.methodology
23 ) 3    GESTATION_THRESHOLD
24 ) 3    Gestation.period.threshold
25 ) 4    HEALTH_STATUS
26 ) 4    Health.status
27 ) 5    DISEASE
28 ) 5    Disease
29 ) 7    CANCER_SITE
```

```
30 ) 7   Cancer.site
31 ) 63   TIME_PERIOD
33 ) 78600   OBS_VALUE
37 ) 5   OBS_STATUS
38 ) 5   Observation.status
39 ) 3   OBS_STATUS2
40 ) 3   Observation.status.2
```

measure에는 어떤 변수가 있는지 찾아보자

```r
status$Measure %>% unique
```

```
 [1] "Self-reported absence from work due to illness"
 [2] "Compensated absence from work due to illness"
 [3] "Preventable mortality"
 [4] "Avoidable mortality"
 [5] "Treatable mortality"
 [6] "Mortality"
 [7] "Injuries in road traffic accidents"
 [8] "Perceived health status"
 [9] "Potential years of life lost"
[10] "Life expectancy"
[11] "Cancer incidence"
[12] "Life expectancy difference (female-male)"
[13] "Life expectancy difference (male-female)"
[14] "Maternal mortality"
[15] "Communicable disease incidence"
[16] "Neonatal mortality"
[17] "Low birthweight"
[18] "Infant mortality"
[19] "Perinatal mortality"
```

1.2.1 Avoidable mortality

이 중에서 Avoidable mortality 만 선택해서 변수들을 살펴보자

```r
status_Am <- status[status$Measure=="Avoidable mortality"&
                      status$Sex=="Total"&
                      status$Unit.of.measure=="Deaths per 100 000 inhabitants",]
checkvar(status_Am)
```

```
5 ) 45   REF_AREA
6 ) 45   Reference.area
31 ) 22   TIME_PERIOD
33 ) 353   OBS_VALUE
37 ) 2   OBS_STATUS
38 ) 2   Observation.status
```

```r
checkvar2(status_Am)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : AVM
10 ) Measure : Avoidable mortality
11 ) UNIT_MEASURE : DT_10P5HB
12 ) Unit.of.measure : Deaths per 100 000 inhabitants
13 ) AGE : _T
14 ) Age : Total
15 ) SEX : _T
16 ) Sex : Total
21 ) CALC_METHODOLOGY : STANDARD
22 ) Calculation.methodology : Standardised rate
39 ) OBS_STATUS2 :
40 ) Observation.status.2 :
```

```r
status_Am$Observation.status %>% unique
```

```
[1] ""                   "Definition differs"
```

```r
status_Am[,c("Reference.area","Observation.status")] %>% table
```

```
                 Observation.status
Reference.area       Definition differs
  Argentina       21                  0
  Australia       21                  0
  Austria         20                  0
  Belgium         19                  0
  Brazil          21                  0
  Bulgaria        16                  0
  Canada          20                  0
  Chile           21                  0
  Colombia        21                  0
  Costa Rica      21                  0
  Croatia         20                  0
  Czechia         22                  0
  Denmark         21                  0
  Estonia         22                  0
  Finland         20                  0
  France          18                  0
  Germany         21                  0
  Greece           7                  0
  Hungary         20                  0
```

```
Iceland         22                   0
Ireland         10                   0
Israel          21                   0
Italy           15                   0
Japan           21                   0
Korea           21                   0
Latvia          22                   0
Lithuania       22                   0
Luxembourg      22                   0
Mexico          21                   0
Netherlands     21                   0
New Zealand     17                   0
Norway          17                   0
Peru            21                   0
Poland          21                   0
Portugal        15                   0
Romania         20                   0
Slovak Republic 18                   0
Slovenia        21                   0
South Africa    19                   0
Spain           22                   0
Sweden          19                   0
Switzerland     21                   0
Türkiye          0                   9
United Kingdom  20                   0
United States   21                   0
```

변수들의 구조가 다차원적인 형태이다. 44개의 변수가 있다면 최대 44차원의 데이터라고 할 수 있다.

실제로는 22개의 변수와 코드 형태로 되어 있어서 22개 차원이고, 또 다시 몇가지 변수는 데이터를 구분하기 위한 태그에 불과하기 때문에 최종적으로는 OBS_VALUE 측정값을 제외하고 다음 7가지 변수가 의미가 있다.

REF_AREA = Reference.area,

MEASURE = Measure,

UNIT_MEASURE = Unit.of.measure,

AGE = Age,

SEX = Sex,

SOCIO_ECON_STATUS = Socio.economic.status

DEATH_CAUSE = Cause.of.death

OBS_VALUE

그런데 우리가 표로 제시해야할 때 사용할 수 있는 변수는 두가지 뿐이다. 대개 시각적으로 쉽게 이해 가능한 차원은 2차원이기 때문이다.

여기서 제시할 도표는 국가별, 연도별 2차원으로 하도록 하자. 그렇게 하려면 나머지 변수들은 1개의 값으로 고정해야 한다. 또 어떤 변수는 여러개의 값을 갖지만 데이터 성격만 구분하는 값이어서 통합하는 것이 나을 수 있다. 이런 경우에는 그 변수를 무시하고 합치면 된다.

```
df_Am <- status[status$Measure=="Avoidable mortality",
                c("Reference.area","Sex","TIME_PERIOD","OBS_VALUE")]
df_Am[,c(3,2,1)] %>% table %>% as.data.frame -> df_table
checkvar(df_table)
```

```
1 ) 22   TIME_PERIOD
2 ) 3   Sex
3 ) 45   Reference.area
4 ) 3   Freq
```

```
df_table[,c(1,2)] %>% table
```

```
           Sex
TIME_PERIOD Female Male Total
       2000     45   45    45
       2001     45   45    45
       2002     45   45    45
       2003     45   45    45
       2004     45   45    45
       2005     45   45    45
       2006     45   45    45
       2007     45   45    45
       2008     45   45    45
       2009     45   45    45
       2010     45   45    45
       2011     45   45    45
       2012     45   45    45
       2013     45   45    45
       2014     45   45    45
       2015     45   45    45
       2016     45   45    45
       2017     45   45    45
       2018     45   45    45
       2019     45   45    45
       2020     45   45    45
       2021     45   45    45
```

```
df_table[,2] %>% unique
```

```
[1] Female Male   Total
Levels: Female Male Total
```

변수들을 분석해본 결과 Sex는 Total로 정하고, (Male, Female 구분할 필요가 있으면 별도로 분석) Unit of measure는 인구 10만명당 비율로 정하기로 하였다. Observation.status는 Turkiye 에서만 Definition 이 달라서 차이가 나지만 큰 상관이 없어서 무시할 수 있다.

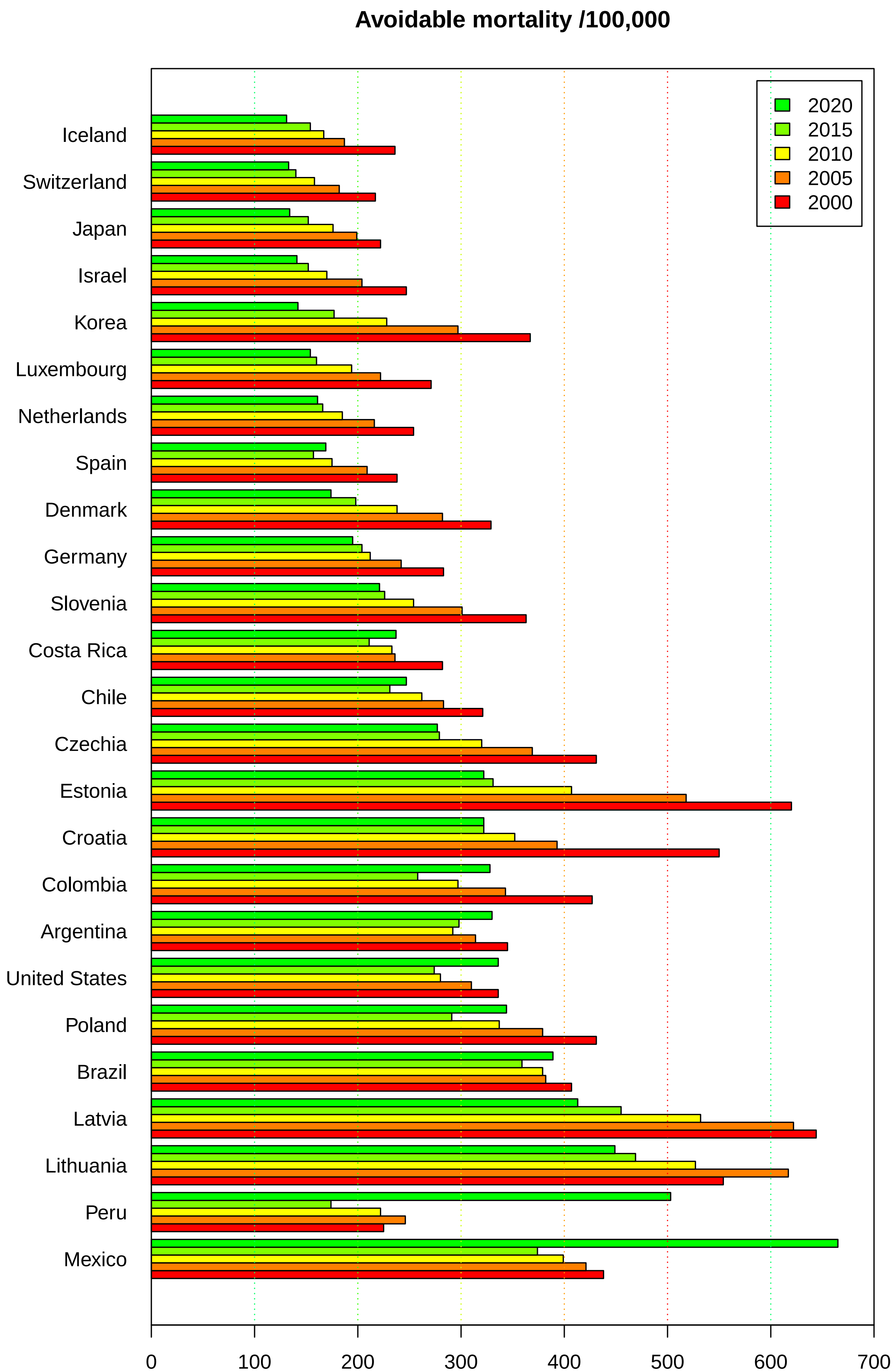
TIME_PERIOD는 2000~2021년까지 있는데 국가별로 중간에 데이터가 없는 연도도 있다. 한국의 경우 2021년 데이터가 없기 때문에 2009, 2014, 2019 이렇게 3개연도를 선택하여 데이터가 전부 다 있는 국가들만 선택한다.

```r
##status_Am[,c("Reference.area","TIME_PERIOD")] %>% table
## 국가별로 연도가 빠진 것이 있는지 check
timetable <- as.data.frame(table(status_Am[status_Am$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

```
    TIME_PERIOD  Reference.area Freq
7          2005       Australia    0
11         2000         Austria    0
20         2020         Belgium    0
26         2000        Bulgaria    0
35         2020          Canada    0
74         2015         Finland    0
80         2020          France    0
86         2000          Greece    0
87         2005          Greece    0
88         2010          Greece    0
95         2020         Hungary    0
101        2000         Ireland    0
102        2005         Ireland    0
105        2020         Ireland    0
111        2000           Italy    0
115        2020           Italy    0
155        2020     New Zealand    0
160        2020          Norway    0
171        2000        Portugal    0
172        2005        Portugal    0
175        2020        Portugal    0
180        2020         Romania    0
184        2015 Slovak Republic    0
185        2020 Slovak Republic    0
195        2020    South Africa    0
205        2020          Sweden    0
211        2000         Türkiye    0
212        2005         Türkiye    0
215        2020         Türkiye    0
216        2000  United Kingdom    0
```

```r
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Am[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Am
status_Am[status_Am$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),] -> status_Am
status_Am[!status_Am$Reference.area%in%excluded,]-> status_Am
status_Am %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Am
mat_Am <- mat_Am[order(-mat_Am$`2020`),] ### 2020년도 사망률 낮은 순서대로 정렬
mat_Am[,2:6]%>% as.matrix %>% t -> matval
```

```
colnames(matval) <- mat_Am$Reference.area
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(12)[1:5],horiz = T, las=1,
                   xlim=c(0,700),
                   main = "Avoidable mortality /100,000",
#                   args.legend = list(x="topright"),
                   legend.text = c(2000,2005,2010,2015,2020) )
abline(v=c(100,200,300,400,500,600), lty=3,col=rainbow(10)[5:1])
box()
```



Avoidable mortality /100,000
2020
2015
2010
2005
2000
Iceland
Switzerland
Japan
Israel
Korea
Luxembourg
Netherlands
Spain
Denmark
Germany
Slovenia
Costa Rica
Chile
Czechia
Estonia
Croatia
Colombia
Argentina
United States
Poland
Brazil
Latvia
Lithuania
Peru
Mexico
0
100
200
300
400
500
600
700

1.2.2 Preventable mortality

이번에는 Avoidable mortality 만 선택해서 변수들을 살펴보자

```r
status_Pm <- status[status$Measure=="Preventable mortality"&
                      status$Sex=="Total"&
                      status$Unit.of.measure=="Deaths per 100 000 inhabitants",]
checkvar(status_Pm)
```

```
5 ) 45   REF_AREA
6 ) 45   Reference.area
31 ) 22   TIME_PERIOD
33 ) 267   OBS_VALUE
37 ) 2   OBS_STATUS
38 ) 2   Observation.status
```

```r
checkvar2(status_Pm)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : PREVM
10 ) Measure : Preventable mortality
11 ) UNIT_MEASURE : DT_10P5HB
12 ) Unit.of.measure : Deaths per 100 000 inhabitants
13 ) AGE : _T
14 ) Age : Total
15 ) SEX : _T
16 ) Sex : Total
21 ) CALC_METHODOLOGY : STANDARD
22 ) Calculation.methodology : Standardised rate
39 ) OBS_STATUS2 :
40 ) Observation.status.2 :
```

```r
status_Pm$Observation.status %>% unique
```

```
[1] ""                    "Definition differs"
```

```r
status_Pm[,c("Reference.area","Observation.status")] %>% table
```

```
                 Observation.status
Reference.area        Definition differs
  Argentina       21                   0
  Australia       21                   0
  Austria         20                   0
```

```
Belgium         19                   0
Brazil          21                   0
Bulgaria        16                   0
Canada          20                   0
Chile           21                   0
Colombia        21                   0
Costa Rica      21                   0
Croatia         20                   0
Czechia         22                   0
Denmark         21                   0
Estonia         22                   0
Finland         20                   0
France          18                   0
Germany         21                   0
Greece           7                   0
Hungary         20                   0
Iceland         22                   0
Ireland         10                   0
Israel          21                   0
Italy           15                   0
Japan           21                   0
Korea           21                   0
Latvia          22                   0
Lithuania       22                   0
Luxembourg      22                   0
Mexico          21                   0
Netherlands     21                   0
New Zealand     17                   0
Norway          17                   0
Peru            21                   0
Poland          21                   0
Portugal        15                   0
Romania         20                   0
Slovak Republic 18                   0
Slovenia        21                   0
South Africa    19                   0
Spain           22                   0
Sweden          19                   0
Switzerland     21                   0
Türkiye          0                   9
United Kingdom  20                   0
United States   21                   0
```

Avoidable mortality와 마찬가지로 Sex는 Total로 정하고, (Male, Female 구분할 필요가 있으면 별도로 분석) Unit of measure는 인구 10만명당 비율로 정하기로 하였다. Observation.status는 Turkiye 에서만 Definition 이 달라서 차이가 나지만 큰 상관이 없어서 무시할 수 있다.

TIME_PERIOD는 2000~2021년까지 있는데 국가별로 중간에 데이터가 없는 연도도 있다. 한국의 경우 2021년 데이터가 없기 때문에 2009, 2014, 2019 이렇게 3개연도를 선택하여 데이터가 전부 다 있는 국가들만 선택한다.

```r
status_Pm[,c("Reference.area","TIME_PERIOD")] %>% table
```

```
                 TIME_PERIOD
Reference.area    2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011
  Argentina          1    1    1    1    1    1    1    1    1    1    1    1
  Australia          1    1    1    1    1    0    1    1    1    1    1    1
  Austria            0    0    1    1    1    1    1    1    1    1    1    1
  Belgium            1    1    1    1    1    1    1    1    1    1    1    1
  Brazil             1    1    1    1    1    1    1    1    1    1    1    1
  Bulgaria           0    0    0    0    0    1    1    1    1    1    1    1
  Canada             1    1    1    1    1    1    1    1    1    1    1    1
  Chile              1    1    1    1    1    1    1    1    1    1    1    1
  Colombia           1    1    1    1    1    1    1    1    1    1    1    1
  Costa Rica         1    1    1    1    1    1    1    1    1    1    1    1
  Croatia            1    1    1    1    1    1    1    1    1    1    1    1
  Czechia            1    1    1    1    1    1    1    1    1    1    1    1
  Denmark            1    1    1    1    1    1    1    1    1    1    1    1
  Estonia            1    1    1    1    1    1    1    1    1    1    1    1
  Finland            1    1    1    1    1    1    1    1    1    1    1    1
  France             1    1    1    1    1    1    1    1    1    1    1    1
  Germany            1    1    1    1    1    1    1    1    1    1    1    1
  Greece             0    0    0    0    0    0    0    0    0    0    0    0
  Hungary            1    1    1    1    1    1    1    1    1    1    1    1
  Iceland            1    1    1    1    1    1    1    1    1    1    1    1
  Ireland            0    0    0    0    0    0    0    1    1    1    1    1
  Israel             1    1    1    1    1    1    1    1    1    1    1    1
  Italy              0    0    0    1    1    1    1    1    1    1    1    1
  Japan              1    1    1    1    1    1    1    1    1    1    1    1
  Korea              1    1    1    1    1    1    1    1    1    1    1    1
  Latvia             1    1    1    1    1    1    1    1    1    1    1    1
  Lithuania          1    1    1    1    1    1    1    1    1    1    1    1
  Luxembourg         1    1    1    1    1    1    1    1    1    1    1    1
  Mexico             1    1    1    1    1    1    1    1    1    1    1    1
  Netherlands        1    1    1    1    1    1    1    1    1    1    1    1
  New Zealand        1    1    1    1    1    1    1    1    1    1    1    1
  Norway             1    1    1    1    1    1    1    1    1    1    1    1
  Peru               1    1    1    1    1    1    1    1    1    1    1    1
  Poland             1    1    1    1    1    1    1    1    1    1    1    1
  Portugal           0    0    1    1    0    0    0    1    1    1    1    1
  Romania            1    1    1    1    1    1    1    1    1    1    1    1
  Slovak Republic    1    1    1    1    1    1    1    1    1    1    1    0
  Slovenia           1    1    1    1    1    1    1    1    1    1    1    1
  South Africa       1    1    1    1    1    1    1    1    1    1    1    1
  Spain              1    1    1    1    1    1    1    1    1    1    1    1
  Sweden             1    1    1    1    1    1    1    1    1    1    1    1
  Switzerland        1    1    1    1    1    1    1    1    1    1    1    1
  Türkiye            0    0    0    0    0    0    0    0    0    1    1    1
  United Kingdom     0    1    1    1    1    1    1    1    1    1    1    1
  United States      1    1    1    1    1    1    1    1    1    1    1    1
```

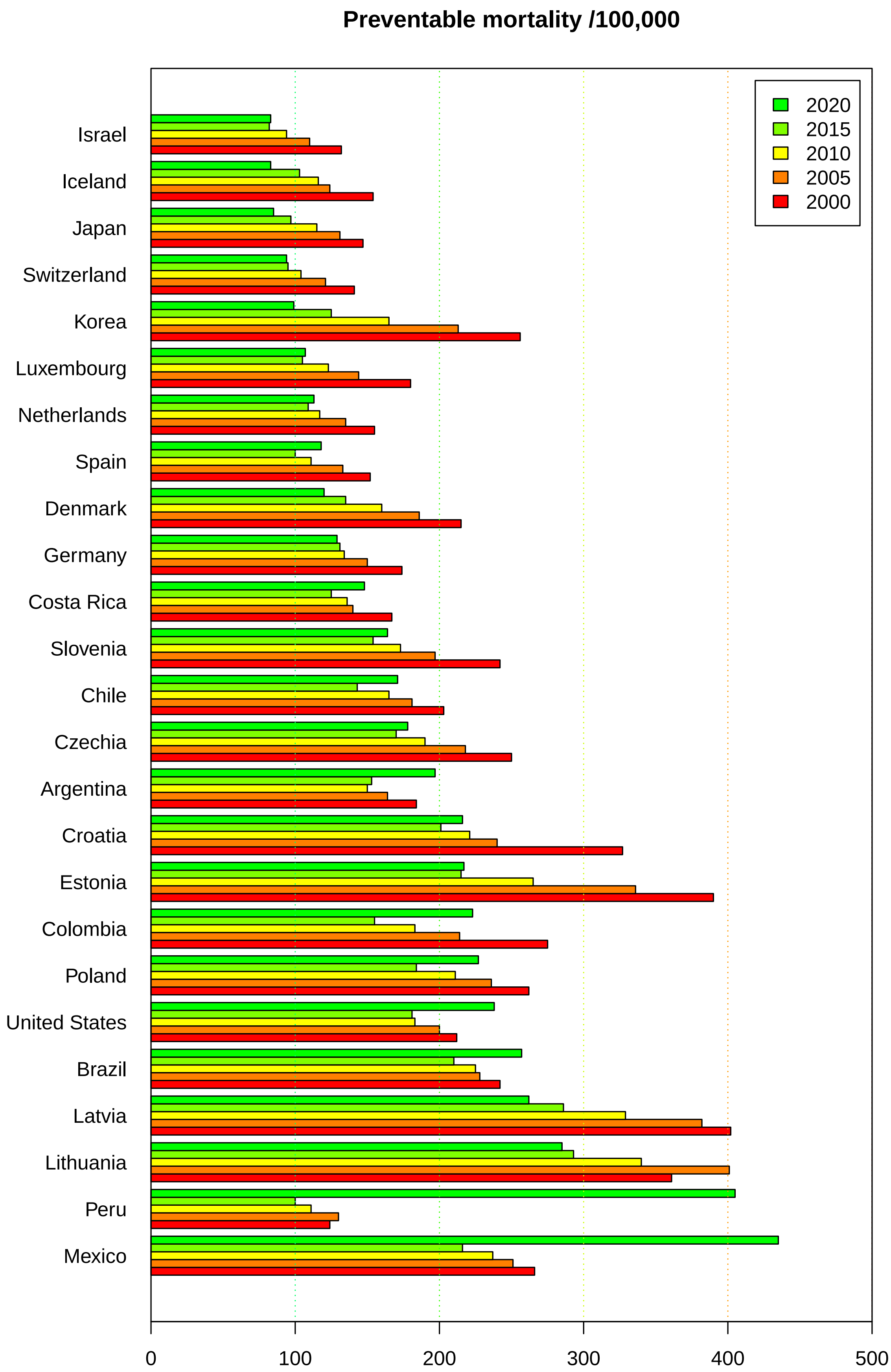
	TIME_PERIOD									
Reference.area	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Argentina	1	1	1	1	1	1	1	1	1	0
Australia	1	1	1	1	1	1	1	1	1	1
Austria	1	1	1	1	1	1	1	1	1	1
Belgium	1	1	1	1	1	1	1	0	0	0
Brazil	1	1	1	1	1	1	1	1	1	0
Bulgaria	1	1	1	1	1	1	1	1	1	0
Canada	1	1	1	1	1	1	1	1	0	0
Chile	1	1	1	1	1	1	1	1	1	0
Colombia	1	1	1	1	1	1	1	1	1	0
Costa Rica	1	1	1	1	1	1	1	1	1	0
Croatia	1	1	1	1	1	1	0	1	1	0
Czechia	1	1	1	1	1	1	1	1	1	1
Denmark	1	1	1	1	1	1	1	1	1	0
Estonia	1	1	1	1	1	1	1	1	1	1
Finland	1	1	1	0	1	1	1	1	1	0
France	1	1	1	1	1	1	0	0	0	0
Germany	1	1	1	1	1	1	1	1	1	0
Greece	0	0	1	1	1	1	1	1	1	0
Hungary	1	1	1	1	1	1	1	1	0	0
Iceland	1	1	1	1	1	1	1	1	1	1
Ireland	1	1	1	1	0	0	1	0	0	0
Israel	1	1	1	1	1	1	1	1	1	0
Italy	1	1	1	1	1	1	0	0	0	0
Japan	1	1	1	1	1	1	1	1	1	0
Korea	1	1	1	1	1	1	1	1	1	0
Latvia	1	1	1	1	1	1	1	1	1	1
Lithuania	1	1	1	1	1	1	1	1	1	1
Luxembourg	1	1	1	1	1	1	1	1	1	1
Mexico	1	1	1	1	1	1	1	1	1	0
Netherlands	1	1	1	1	1	1	1	1	1	0
New Zealand	1	1	1	1	1	0	0	0	0	0
Norway	1	1	1	1	1	0	0	0	0	0
Peru	1	1	1	1	1	1	1	1	1	0
Poland	1	1	1	1	1	1	1	1	1	0
Portugal	1	1	1	1	1	1	1	1	0	0
Romania	1	1	1	1	1	1	1	1	0	0
Slovak Republic	1	1	1	0	1	1	1	1	0	0
Slovenia	1	1	1	1	1	1	1	1	1	0
South Africa	1	1	1	1	1	1	1	0	0	0
Spain	1	1	1	1	1	1	1	1	1	1
Sweden	1	1	1	1	1	1	1	0	0	0
Switzerland	1	1	1	1	1	1	1	1	1	0
Türkiye	1	1	1	1	1	0	0	1	0	0
United Kingdom	1	1	1	1	1	1	1	1	1	0
United States	1	1	1	1	1	1	1	1	1	0

```r
## 국가별로 연도가 빠진 것이 있는지 check
timetable <- as.data.frame(table(status_Pm[status_Pm$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

```
    TIME_PERIOD  Reference.area Freq
7          2005       Australia    0
11         2000         Austria    0
20         2020         Belgium    0
26         2000        Bulgaria    0
35         2020          Canada    0
74         2015         Finland    0
80         2020          France    0
86         2000          Greece    0
87         2005          Greece    0
88         2010          Greece    0
95         2020         Hungary    0
101        2000         Ireland    0
102        2005         Ireland    0
105        2020         Ireland    0
111        2000           Italy    0
115        2020           Italy    0
155        2020     New Zealand    0
160        2020          Norway    0
171        2000        Portugal    0
172        2005        Portugal    0
175        2020        Portugal    0
180        2020         Romania    0
184        2015 Slovak Republic    0
185        2020 Slovak Republic    0
195        2020    South Africa    0
205        2020          Sweden    0
211        2000         Türkiye    0
212        2005         Türkiye    0
215        2020         Türkiye    0
216        2000  United Kingdom    0
```

```r
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Pm[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Pm
status_Pm[status_Pm$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),] -> status_Pm
status_Pm[!status_Pm$Reference.area%in%excluded,]-> status_Pm
status_Pm %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Pm
mat_Pm <- mat_Pm[order(-mat_Pm$`2020`),] ### 2020년도 사망률 낮은 순서대로 정렬
mat_Pm[,2:6]%>% as.matrix %>% t -> matval
colnames(matval) <- mat_Pm$Reference.area
```

```
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(12)[1:5],horiz = T, las=1,
                   xlim=c(0,500),
                   main = "Preventable mortality /100,000",
#                   args.legend = list(x="topright"),
                   legend.text = c(2000,2005,2010,2015,2020) )
abline(v=c(100,200,300,400,500,600), lty=3,col=rainbow(10)[5:1])
box()
```


Preventable mortality /100,000
2020
2015
2010
2005
2000
Israel
Iceland
Japan
Switzerland
Korea
Luxembourg
Netherlands
Spain
Denmark
Germany
Costa Rica
Slovenia
Chile
Czechia
Argentina
Croatia
Estonia
Colombia
Poland
United States
Brazil
Latvia
Lithuania
Peru
Mexico
0
100
200
300
400
500

1.2.3 Treatable mortality

이번에는 Treatable mortality 만 선택해서 변수들을 살펴보자

```
status$Measure %>% unique
```

```
 [1] "Self-reported absence from work due to illness"
 [2] "Compensated absence from work due to illness"
 [3] "Preventable mortality"
 [4] "Avoidable mortality"
 [5] "Treatable mortality"
 [6] "Mortality"
 [7] "Injuries in road traffic accidents"
 [8] "Perceived health status"
 [9] "Potential years of life lost"
[10] "Life expectancy"
[11] "Cancer incidence"
[12] "Life expectancy difference (female-male)"
[13] "Life expectancy difference (male-female)"
[14] "Maternal mortality"
[15] "Communicable disease incidence"
[16] "Neonatal mortality"
[17] "Low birthweight"
[18] "Infant mortality"
[19] "Perinatal mortality"
```

Measure 변수들 중에서 선택하고 있다.

```
status_Tm <- status[status$Measure=="Treatable mortality"&
                      status$Sex=="Total"&
                      status$Unit.of.measure=="Deaths per 100 000 inhabitants",]
checkvar(status_Tm)
```

```
5 ) 45   REF_AREA
6 ) 45   Reference.area
31 ) 22   TIME_PERIOD
33 ) 194   OBS_VALUE
37 ) 2   OBS_STATUS
38 ) 2   Observation.status
```

```
checkvar2(status_Tm)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
```

```
9 ) MEASURE : TRTM
10 ) Measure : Treatable mortality
11 ) UNIT_MEASURE : DT_10P5HB
12 ) Unit.of.measure : Deaths per 100 000 inhabitants
13 ) AGE : _T
14 ) Age : Total
15 ) SEX : _T
16 ) Sex : Total
21 ) CALC_METHODOLOGY : STANDARD
22 ) Calculation.methodology : Standardised rate
39 ) OBS_STATUS2 :
40 ) Observation.status.2 :
```

```r
### status_Tm$Observation.status %>% unique
### status_Tm[,c("Reference.area","Observation.status")] %>% table
```

Avoidable mortality와 마찬가지로 Sex는 Total로 정하고, Unit of measure는 인구 10만명당 비율로 정한다. Observation.status는 Turkiye 에서만 Definition 이 달라서 차이가 나지만 큰 상관이 없어서 무시할 수 있다. TIME_PERIOD는 2000~2021년까지 있는데 국가별로 중간에 데이터가 없는 연도도 있다.

```r
status_Tm[,c("Reference.area","TIME_PERIOD")] %>% table
```

Reference.area \ TIME_PERIOD	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
Argentina	1	1	1	1	1	1	1	1	1	1	1	1
Australia	1	1	1	1	1	0	1	1	1	1	1	1
Austria	0	0	1	1	1	1	1	1	1	1	1	1
Belgium	1	1	1	1	1	1	1	1	1	1	1	1
Brazil	1	1	1	1	1	1	1	1	1	1	1	1
Bulgaria	0	0	0	0	0	1	1	1	1	1	1	1
Canada	1	1	1	1	1	1	1	1	1	1	1	1
Chile	1	1	1	1	1	1	1	1	1	1	1	1
Colombia	1	1	1	1	1	1	1	1	1	1	1	1
Costa Rica	1	1	1	1	1	1	1	1	1	1	1	1
Croatia	1	1	1	1	1	1	1	1	1	1	1	1
Czechia	1	1	1	1	1	1	1	1	1	1	1	1
Denmark	1	1	1	1	1	1	1	1	1	1	1	1
Estonia	1	1	1	1	1	1	1	1	1	1	1	1
Finland	1	1	1	1	1	1	1	1	1	1	1	1
France	1	1	1	1	1	1	1	1	1	1	1	1
Germany	1	1	1	1	1	1	1	1	1	1	1	1
Greece	0	0	0	0	0	0	0	0	0	0	0	0
Hungary	1	1	1	1	1	1	1	1	1	1	1	1
Iceland	1	1	1	1	1	1	1	1	1	1	1	1
Ireland	0	0	0	0	0	0	0	1	1	1	1	1
Israel	1	1	1	1	1	1	1	1	1	1	1	1
Italy	0	0	0	1	1	1	1	1	1	1	1	1
Japan	1	1	1	1	1	1	1	1	1	1	1	1

Korea	1	1	1	1	1	1	1	1	1	1	1	1
Latvia	1	1	1	1	1	1	1	1	1	1	1	1
Lithuania	1	1	1	1	1	1	1	1	1	1	1	1
Luxembourg	1	1	1	1	1	1	1	1	1	1	1	1
Mexico	1	1	1	1	1	1	1	1	1	1	1	1
Netherlands	1	1	1	1	1	1	1	1	1	1	1	1
New Zealand	1	1	1	1	1	1	1	1	1	1	1	1
Norway	1	1	1	1	1	1	1	1	1	1	1	1
Peru	1	1	1	1	1	1	1	1	1	1	1	1
Poland	1	1	1	1	1	1	1	1	1	1	1	1
Portugal	0	0	1	1	0	0	0	1	1	1	1	1
Romania	1	1	1	1	1	1	1	1	1	1	1	1
Slovak Republic	1	1	1	1	1	1	1	1	1	1	1	0
Slovenia	1	1	1	1	1	1	1	1	1	1	1	1
South Africa	1	1	1	1	1	1	1	1	1	1	1	1
Spain	1	1	1	1	1	1	1	1	1	1	1	1
Sweden	1	1	1	1	1	1	1	1	1	1	1	1
Switzerland	1	1	1	1	1	1	1	1	1	1	1	1
Türkiye	0	0	0	0	0	0	0	0	0	1	1	1
United Kingdom	0	1	1	1	1	1	1	1	1	1	1	1
United States	1	1	1	1	1	1	1	1	1	1	1	1

	TIME_PERIOD									
Reference.area	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Argentina	1	1	1	1	1	1	1	1	1	0
Australia	1	1	1	1	1	1	1	1	1	1
Austria	1	1	1	1	1	1	1	1	1	1
Belgium	1	1	1	1	1	1	1	0	0	0
Brazil	1	1	1	1	1	1	1	1	1	0
Bulgaria	1	1	1	1	1	1	1	1	1	0
Canada	1	1	1	1	1	1	1	1	0	0
Chile	1	1	1	1	1	1	1	1	1	0
Colombia	1	1	1	1	1	1	1	1	1	0
Costa Rica	1	1	1	1	1	1	1	1	1	0
Croatia	1	1	1	1	1	1	0	1	1	0
Czechia	1	1	1	1	1	1	1	1	1	1
Denmark	1	1	1	1	1	1	1	1	1	0
Estonia	1	1	1	1	1	1	1	1	1	1
Finland	1	1	1	0	1	1	1	1	1	0
France	1	1	1	1	1	1	0	0	0	0
Germany	1	1	1	1	1	1	1	1	1	0
Greece	0	0	1	1	1	1	1	1	1	0
Hungary	1	1	1	1	1	1	1	1	0	0
Iceland	1	1	1	1	1	1	1	1	1	1
Ireland	1	1	1	1	0	0	1	0	0	0
Israel	1	1	1	1	1	1	1	1	1	0
Italy	1	1	1	1	1	1	0	0	0	0
Japan	1	1	1	1	1	1	1	1	1	0
Korea	1	1	1	1	1	1	1	1	1	0
Latvia	1	1	1	1	1	1	1	1	1	1

```
  Lithuania          1    1    1    1    1    1    1    1    1    1
  Luxembourg         1    1    1    1    1    1    1    1    1    1
  Mexico             1    1    1    1    1    1    1    1    1    0
  Netherlands        1    1    1    1    1    1    1    1    1    0
  New Zealand        1    1    1    1    1    0    0    0    0    0
  Norway             1    1    1    1    1    0    0    0    0    0
  Peru               1    1    1    1    1    1    1    1    1    0
  Poland             1    1    1    1    1    1    1    1    1    0
  Portugal           1    1    1    1    1    1    1    1    0    0
  Romania            1    1    1    1    1    1    1    1    0    0
  Slovak Republic    1    1    1    0    1    1    1    1    0    0
  Slovenia           1    1    1    1    1    1    1    1    1    0
  South Africa       1    1    1    1    1    1    1    0    0    0
  Spain              1    1    1    1    1    1    1    1    1    1
  Sweden             1    1    1    1    1    1    1    0    0    0
  Switzerland        1    1    1    1    1    1    1    1    1    0
  Türkiye            1    1    1    1    1    0    0    1    0    0
  United Kingdom     1    1    1    1    1    1    1    1    1    0
  United States      1    1    1    1    1    1    1    1    1    0
```

```r
## 국가별로 연도가 빠진 것이 있는지 check
timetable <- as.data.frame(table(status_Tm[status_Pm$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

```
    TIME_PERIOD  Reference.area Freq
22         2021       Argentina    0
28         2005       Australia    0
45         2000         Austria    0
46         2001         Austria    0
86         2019         Belgium    0
87         2020         Belgium    0
88         2021         Belgium    0
110        2021          Brazil    0
111        2000        Bulgaria    0
112        2001        Bulgaria    0
113        2002        Bulgaria    0
114        2003        Bulgaria    0
115        2004        Bulgaria    0
132        2021        Bulgaria    0
153        2020          Canada    0
154        2021          Canada    0
176        2021           Chile    0
198        2021        Colombia    0
220        2021      Costa Rica    0
239        2018         Croatia    0
242        2021         Croatia    0
286        2021         Denmark    0
```

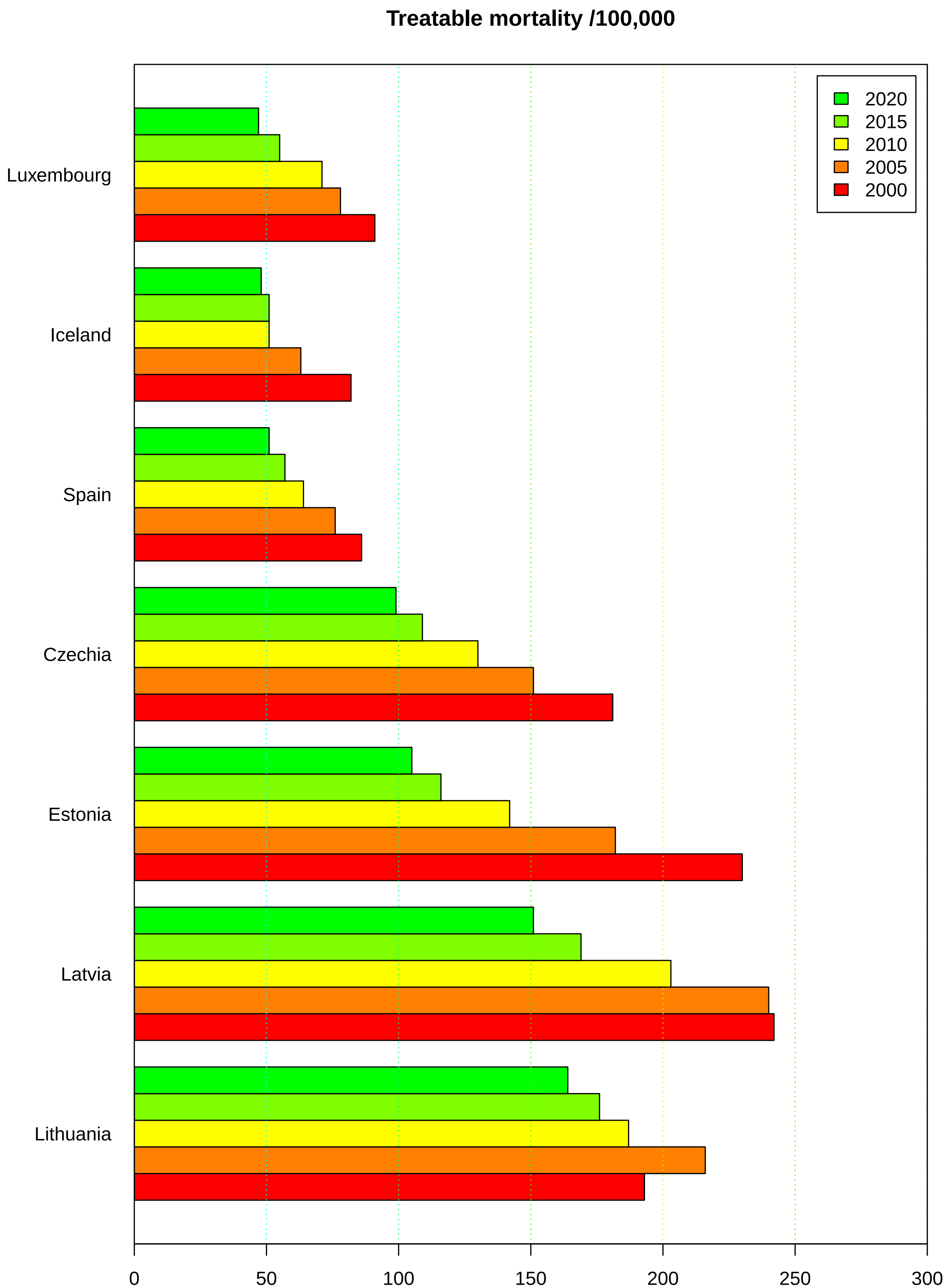
324	2015	Finland	0
330	2021	Finland	0
349	2018	France	0
350	2019	France	0
351	2020	France	0
352	2021	France	0
374	2021	Germany	0
375	2000	Greece	0
376	2001	Greece	0
377	2002	Greece	0
378	2003	Greece	0
379	2004	Greece	0
380	2005	Greece	0
381	2006	Greece	0
382	2007	Greece	0
383	2008	Greece	0
384	2009	Greece	0
385	2010	Greece	0
386	2011	Greece	0
387	2012	Greece	0
388	2013	Greece	0
396	2021	Greece	0
417	2020	Hungary	0
418	2021	Hungary	0
441	2000	Ireland	0
442	2001	Ireland	0
443	2002	Ireland	0
444	2003	Ireland	0
445	2004	Ireland	0
446	2005	Ireland	0
447	2006	Ireland	0
457	2016	Ireland	0
458	2017	Ireland	0
460	2019	Ireland	0
461	2020	Ireland	0
462	2021	Ireland	0
484	2021	Israel	0
485	2000	Italy	0
486	2001	Italy	0
487	2002	Italy	0
503	2018	Italy	0
504	2019	Italy	0
505	2020	Italy	0
506	2021	Italy	0
528	2021	Japan	0
550	2021	Korea	0
638	2021	Mexico	0
660	2021	Netherlands	0
678	2017	New Zealand	0

679	2018	New Zealand	0
680	2019	New Zealand	0
681	2020	New Zealand	0
682	2021	New Zealand	0
700	2017	Norway	0
701	2018	Norway	0
702	2019	Norway	0
703	2020	Norway	0
704	2021	Norway	0
726	2021	Peru	0
748	2021	Poland	0
749	2000	Portugal	0
750	2001	Portugal	0
753	2004	Portugal	0
754	2005	Portugal	0
755	2006	Portugal	0
769	2020	Portugal	0
770	2021	Portugal	0
791	2020	Romania	0
792	2021	Romania	0
804	2011	Slovak Republic	0
808	2015	Slovak Republic	0
813	2020	Slovak Republic	0
814	2021	Slovak Republic	0
836	2021	Slovenia	0
856	2019	South Africa	0
857	2020	South Africa	0
858	2021	South Africa	0
900	2019	Sweden	0
901	2020	Sweden	0
902	2021	Sweden	0
924	2021	Switzerland	0
925	2000	Türkiye	0
926	2001	Türkiye	0
927	2002	Türkiye	0
928	2003	Türkiye	0
929	2004	Türkiye	0
930	2005	Türkiye	0
931	2006	Türkiye	0
932	2007	Türkiye	0
933	2008	Türkiye	0
942	2017	Türkiye	0
943	2018	Türkiye	0
945	2020	Türkiye	0
946	2021	Türkiye	0
947	2000	United Kingdom	0
968	2021	United Kingdom	0
990	2021	United States	0

```
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Tm[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Tm
status_Tm[status_Tm$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),] -> status_Tm
status_Tm[!status_Tm$Reference.area%in%excluded,]-> status_Tm
status_Tm %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Tm
mat_Tm <- mat_Tm[order(-mat_Tm$`2020`),] ### 2020년도 사망률 낮은 순서대로 정렬
mat_Tm %>% str
```

```
'data.frame':   7 obs. of  6 variables:
 $ Reference.area: chr  "Lithuania" "Latvia" "Estonia" "Czechia" ...
 $ 2000          : num  193 242 230 181 86 82 91
 $ 2005          : num  216 240 182 151 76 63 78
 $ 2010          : num  187 203 142 130 64 51 71
 $ 2015          : num  176 169 116 109 57 51 55
 $ 2020          : num  164 151 105 99 51 48 47
```

```
mat_Tm[,2:6]%>% as.matrix %>% t -> matval
colnames(matval) <- mat_Tm$Reference.area
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(12)[1:5],horiz = T, las=1,
                   xlim=c(0,300),
                   main = "Treatable mortality /100,000",
#                   args.legend = list(x="topright"),
                   legend.text = c(2000,2005,2010,2015,2020) )+
abline(v=c(50,100,150,200,250,300), lty=3,col=rainbow(10)[6:1])+
box()
```


Treatable mortality /100,000
2020
2015
2010
2005
2000
Luxembourg
Iceland
Spain
Czechia
Estonia
Latvia
Lithuania
0
50
100
150
200
250
300

```
numeric(0)
```

1.2.4 Perceived health status

Perceived health status 에 대한 변수들을 살펴보자

```
status_Phs <- status[status$Measure=="Perceived health status"&
                        status$Sex=="Total",]
checkvar(status_Phs)
```

```
5 ) 41   REF_AREA
6 ) 41   Reference.area
13 ) 5   AGE
14 ) 5   Age
17 ) 6   SOCIO_ECON_STATUS
18 ) 6   Socio.economic.status
25 ) 3   HEALTH_STATUS
26 ) 3   Health.status
31 ) 43   TIME_PERIOD
33 ) 962   OBS_VALUE
37 ) 3   OBS_STATUS
38 ) 3   Observation.status
39 ) 2   OBS_STATUS2
40 ) 2   Observation.status.2
```

```
checkvar2(status_Phs)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : PHS
10 ) Measure : Perceived health status
11 ) UNIT_MEASURE : PT_POP_AGE
12 ) Unit.of.measure : Percentage of population in the same age
15 ) SEX : _T
16 ) Sex : Total
21 ) CALC_METHODOLOGY : CRUDE
22 ) Calculation.methodology : Crude rate
```

```
status_Phs[,"Observation.status"] %>% unique
```

```
[1] ""                    "Time series break"  "Definition differs"
```

```
status_Phs[,c("SOCIO_ECON_STATUS","HEALTH_STATUS") ] %>% table
```

```
                 HEALTH_STATUS
SOCIO_ECON_STATUS    B    F    G
      _Z           638  623 3627
      ISCED11_0T2    0    0  627
      ISCED11_3T4    0    0  627
      ISCED11_5T8    0    0  627
      Q1             0    0  623
      Q5             0    0  623
```

```
status_Phs[,c("Socio.economic.status","SOCIO_ECON_STATUS") ] %>% table
```

```
                                                                SOCIO_ECON_STATUS
Socio.economic.status                                              _Z
  Income quintile 1 (lowest)                                        0
  Income quintile 5 (highest)                                       0
  Not applicable                                                 4888
  Pre-primary, primary and lower secondary education                0
  Tertiary education                                                0
  Upper secondary and post-secondary non-tertiary all programmes    0
                                                                SOCIO_ECON_STATUS
Socio.economic.status                                             ISCED11_0T2
  Income quintile 1 (lowest)                                                0
  Income quintile 5 (highest)                                               0
  Not applicable                                                            0
  Pre-primary, primary and lower secondary education                      627
  Tertiary education                                                        0
  Upper secondary and post-secondary non-tertiary all programmes            0
                                                                SOCIO_ECON_STATUS
Socio.economic.status                                             ISCED11_3T4
  Income quintile 1 (lowest)                                                0
  Income quintile 5 (highest)                                               0
  Not applicable                                                            0
  Pre-primary, primary and lower secondary education                        0
  Tertiary education                                                        0
  Upper secondary and post-secondary non-tertiary all programmes          627
                                                                SOCIO_ECON_STATUS
Socio.economic.status                                             ISCED11_5T8
  Income quintile 1 (lowest)                                                0
  Income quintile 5 (highest)                                               0
  Not applicable                                                            0
  Pre-primary, primary and lower secondary education                        0
  Tertiary education                                                      627
  Upper secondary and post-secondary non-tertiary all programmes            0
                                                                SOCIO_ECON_STATUS
Socio.economic.status                                               Q1  Q5
  Income quintile 1 (lowest)                                       623   0
  Income quintile 5 (highest)                                        0 623
  Not applicable                                                     0   0
  Pre-primary, primary and lower secondary education                 0   0
```

```
  Tertiary education                                               0    0
  Upper secondary and post-secondary non-tertiary all programmes    0    0
```

```r
status_Phs[,c("Health.status","HEALTH_STATUS") ] %>% table
```

```
                                 HEALTH_STATUS
Health.status                        B    F    G
  Bad/very bad health              638    0    0
  Fair (not good, not bad) health    0  623    0
  Good/very good health              0    0 6754
```

```r
status_Phs[,c("Health.status","REF_AREA") ] %>% table
```

```
                                 REF_AREA
Health.status                      AUS AUT BEL BGR CAN CHE CHL COL CRI CZE DEU
  Bad/very bad health                6  19  19  15  18  15   7   2   1  18  18
  Fair (not good, not bad) health    6  19  19  15  18  15   7   2   1  18  18
  Good/very good health             58 183 191 149 163 156  76  20   5 189 179
                                 REF_AREA
Health.status                      DNK ESP EST FIN FRA GBR GRC HRV HUN IRL ISL
  Bad/very bad health               19  19  19  19  19  15  19  13  18  19  15
  Fair (not good, not bad) health   19  19  19  19  19  15  19  13  18  19  15
  Good/very good health            183 207 183 255 183 208 187 120 181 191 155
                                 REF_AREA
Health.status                      ISR ITA JPN KOR LTU LUX LVA MEX NLD NOR NZL
  Bad/very bad health               20  22   7  10  18  19  18   0  18  17  12
  Fair (not good, not bad) health    5  22   7  10  18  19  18   0  18  17  12
  Good/very good health            200 202  86 109 173 187 173  25 261 177 134
                                 REF_AREA
Health.status                      POL PRT ROU SVK SVN SWE TUR USA
  Bad/very bad health               18  19  15  18  18  19  16  22
  Fair (not good, not bad) health   18  19  15  18  18  19  16  22
  Good/very good health            185 183 149 173 173 299 165 278
```

```r
status_Phs[,c("TIME_PERIOD","SOCIO_ECON_STATUS") ] %>% table
```

```
           SOCIO_ECON_STATUS
TIME_PERIOD  _Z ISCED11_0T2 ISCED11_3T4 ISCED11_5T8  Q1  Q5
       1980  11           0           0           0   0   0
       1981   7           0           0           0   0   0
       1982  11           0           0           0   0   0
       1983  15           0           0           0   0   0
       1984  19           0           0           0   0   0
       1985  15           0           0           0   0   0
       1986  23           0           0           0   0   0
       1987  19           0           0           0   0   0
       1988  15           0           0           0   0   0
       1989  27           0           0           0   0   0
```

```
     1990  15           0           0           0   0   0
     1991  19           0           0           0   0   0
     1992  31           1           1           1   0   0
     1993  27           0           0           0   0   0
     1994  27           0           0           0   0   0
     1995  39           1           1           1   0   0
     1996  31           0           0           0   0   0
     1997  39           0           0           0   0   0
     1998  43           0           0           0   0   0
     1999  31           1           1           1   0   0
     2000  46           3           3           3   2   2
     2001  68           2           2           2   2   2
     2002  61           3           3           3   2   2
     2003  78           6           6           6   4   4
     2004 145          17          17          17  18  18
     2005 200          27          27          27  28  28
     2006 214          29          29          29  29  29
     2007 230          30          30          30  34  34
     2008 230          32          32          32  32  32
     2009 230          32          32          32  33  33
     2010 244          33          33          33  34  34
     2011 236          34          34          34  35  35
     2012 243          35          35          35  34  34
     2013 250          36          36          36  36  36
     2014 251          36          36          36  35  35
     2015 244          35          35          35  35  35
     2016 222          36          36          36  35  35
     2017 223          36          36          36  36  36
     2018 230          36          36          36  35  35
     2019 218          35          35          35  35  35
     2020 199          32          32          32  31  31
     2021 198          32          32          32  32  32
     2022 164          27          27          27  26  26
```

```
status_Phs[,c("REF_AREA","SOCIO_ECON_STATUS") ] %>% table
```

```
        SOCIO_ECON_STATUS
REF_AREA   _Z ISCED11_0T2 ISCED11_3T4 ISCED11_5T8  Q1  Q5
     AUS   54           2           2           2   5   5
     AUT  126          19          19          19  19  19
     BEL  134          19          19          19  19  19
     BGR   98          15          15          15  18  18
     CAN  138          11          11          11  14  14
     CHE  111          15          15          15  15  15
     CHL   50           8           8           8   8   8
     COL   14           2           2           2   2   2
     CRI    7           0           0           0   0   0
     CZE  135          18          18          18  18  18
     DEU  125          18          18          18  18  18
```

```
    DNK 126          19          19          19  19  19
    ESP 150          19          19          19  19  19
    EST 126          19          19          19  19  19
    FIN 198          19          19          19  19  19
    FRA 126          19          19          19  19  19
    GBR 163          15          15          15  15  15
    GRC 130          19          19          19  19  19
    HRV  81          13          13          13  13  13
    HUN 127          18          18          18  18  18
    IRL 134          19          19          19  19  19
    ISL 110          15          15          15  15  15
    ISR 125          20          20          20  20  20
    ITA 156          18          18          18  18  18
    JPN  77           3           3           3   7   7
    KOR  90          13          13          13   0   0
    LTU 119          18          18          18  18  18
    LUX 130          19          19          19  19  19
    LVA 119          18          18          18  18  18
    MEX  25           0           0           0   0   0
    NLD 207          18          18          18  18  18
    NOR 126          17          17          17  17  17
    NZL  93          13          13          13  13  13
    POL 131          18          18          18  18  18
    PRT 126          19          19          19  19  19
    ROU  98          15          15          15  18  18
    SVK 119          18          18          18  18  18
    SVN 119          18          18          18  18  18
    SWE 222          23          23          23  23  23
    TUR 117          16          16          16  16  16
    USA 226          22          22          22  15  15
```

```
status_Phs[,c("Age","HEALTH_STATUS") ] %>% table
```

```
                     HEALTH_STATUS
Age                      B    F    G
  15 years or over     638  623 3778
  65 years or over       0    0  775
  From 15 to 24 years    0    0  799
  From 25 to 44 years    0    0  604
  From 45 to 64 years    0    0  798
```

```
status_Phs[,c("Age","SOCIO_ECON_STATUS") ] %>% table
```

```
                     SOCIO_ECON_STATUS
Age                     _Z ISCED11_0T2 ISCED11_3T4 ISCED11_5T8   Q1   Q5
  15 years or over    1912         627         627         627  623  623
  65 years or over     775           0           0           0    0    0
  From 15 to 24 years  799           0           0           0    0    0
```

```
  From 25 to 44 years  604           0            0            0     0     0
  From 45 to 64 years  798           0            0            0     0     0
```

이 부분은 연령대 구분이 문제. 연령별로 세분하면 좋을 것 같지만 전체적인 것만 평균적으로 보기 위해서 한국을 대상으로, 연령은 15 years or over로 한정하도록 해서 데이터를 탐색해 본다.

```
temp_Phs <- status_Phs[status_Phs$Age=="15 years or over"&
                       status_Phs$REF_AREA=="KOR",]
temp_Phs[,c("TIME_PERIOD","HEALTH_STATUS","SOCIO_ECON_STATUS") ] %>% table
```

```
, , SOCIO_ECON_STATUS = _Z

           HEALTH_STATUS
TIME_PERIOD B F G
       1992 0 0 0
       1995 0 0 0
       1999 0 0 0
       2003 1 1 1
       2006 1 1 1
       2008 1 1 1
       2010 1 1 1
       2012 1 1 1
       2014 1 1 1
       2016 1 1 1
       2018 1 1 1
       2020 1 1 1
       2022 1 1 1

, , SOCIO_ECON_STATUS = ISCED11_0T2

           HEALTH_STATUS
TIME_PERIOD B F G
       1992 0 0 1
       1995 0 0 1
       1999 0 0 1
       2003 0 0 1
       2006 0 0 1
       2008 0 0 1
       2010 0 0 1
       2012 0 0 1
       2014 0 0 1
       2016 0 0 1
       2018 0 0 1
       2020 0 0 1
       2022 0 0 1

, , SOCIO_ECON_STATUS = ISCED11_3T4
```

```
          HEALTH_STATUS
TIME_PERIOD B F G
       1992 0 0 1
       1995 0 0 1
       1999 0 0 1
       2003 0 0 1
       2006 0 0 1
       2008 0 0 1
       2010 0 0 1
       2012 0 0 1
       2014 0 0 1
       2016 0 0 1
       2018 0 0 1
       2020 0 0 1
       2022 0 0 1

, , SOCIO_ECON_STATUS = ISCED11_5T8

          HEALTH_STATUS
TIME_PERIOD B F G
       1992 0 0 1
       1995 0 0 1
       1999 0 0 1
       2003 0 0 1
       2006 0 0 1
       2008 0 0 1
       2010 0 0 1
       2012 0 0 1
       2014 0 0 1
       2016 0 0 1
       2018 0 0 1
       2020 0 0 1
       2022 0 0 1
```

health status가 good 인 경우의 데이터는 어떤 상태인지 확인해보자

```r
temp_Phs <- status_Phs[status_Phs$Age=="15 years or over"&
                       status_Phs$REF_AREA=="KOR"&
                       status_Phs$HEALTH_STATUS=="G",]
temp_Phs[,c("TIME_PERIOD","SOCIO_ECON_STATUS","OBS_VALUE") ] %>%
  spread(key=SOCIO_ECON_STATUS, value = OBS_VALUE)
```

```
  TIME_PERIOD   _Z ISCED11_0T2 ISCED11_3T4 ISCED11_5T8
1        1992   NA        39.3        52.5        56.4
2        1995   NA        37.0        48.4        50.7
3        1999   NA        33.9        47.4        52.1
4        2003 42.9        31.2        46.4        53.7
5        2006 44.6        32.2        47.8        54.9
```

```
6          2008 51.6         35.9        55.3        62.7
7          2010 46.8         33.8        48.8        56.5
8          2012 44.3         31.6        46.0        53.4
9          2014 47.7         33.0        48.6        58.0
10         2016 46.3         33.3        46.4        55.4
11         2018 48.1         33.0        48.3        57.2
12         2020 49.6         32.2        50.0        59.3
13         2022 52.4         33.8        53.0        61.3
```

Socio.economic.status 는 Not applicable 한가지로 통일해도 좋을 것 같다.

```r
status_Phs <- status_Phs[status_Phs$Age=="15 years or over"&
                       status_Phs$Socio.economic.status=="Not applicable"&
                       status_Phs$HEALTH_STATUS=="G" ,]
```

```r
status_Phs[,c("Reference.area","TIME_PERIOD")] %>% table
```

```
                 TIME_PERIOD
Reference.area    2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011
  Australia          0    1    0    0    1    0    0    1    0    0    0    1
  Austria            0    0    0    0    1    1    1    1    1    1    1    1
  Belgium            0    0    0    0    1    1    1    1    1    1    1    1
  Bulgaria           0    0    0    0    0    0    0    0    1    1    1    1
  Canada             0    1    0    1    0    1    0    1    1    1    1    1
  Chile              1    0    0    0    0    0    1    0    0    1    0    0
  Colombia           0    0    0    0    0    0    0    0    0    0    0    0
  Costa Rica         0    0    0    0    0    0    0    0    0    0    0    0
  Croatia            0    0    0    0    0    0    0    0    0    0    1    1
  Czechia            0    0    0    0    0    1    1    1    1    1    1    1
  Denmark            0    0    0    0    1    1    1    1    1    1    1    1
  Estonia            0    0    0    0    1    1    1    1    1    1    1    1
  Finland            0    0    0    0    1    1    1    1    1    1    1    1
  France             0    0    0    0    1    1    1    1    1    1    1    1
  Germany            0    0    0    0    0    1    1    1    1    1    1    1
  Greece             0    0    0    0    1    1    1    1    1    1    1    1
  Hungary            0    0    0    0    0    1    1    1    1    1    1    1
  Iceland            0    0    0    0    1    1    1    1    1    1    1    1
  Ireland            0    0    0    0    1    1    1    1    1    1    1    1
  Israel             0    0    1    1    1    1    1    1    1    1    1    1
  Italy              1    1    1    1    1    1    1    1    1    1    1    1
  Japan              0    1    0    0    1    0    0    1    0    0    1    0
  Korea              0    0    0    1    0    0    1    0    1    0    1    0
  Latvia             0    0    0    0    0    1    1    1    1    1    1    1
  Lithuania          0    0    0    0    0    1    1    1    1    1    1    1
  Luxembourg         0    0    0    0    1    1    1    1    1    1    1    1
  Mexico             1    1    1    0    0    1    1    0    0    0    0    0
  Netherlands        0    0    0    0    0    1    1    1    1    1    1    1
  New Zealand        0    0    0    1    0    0    0    1    0    0    0    0
```

Norway	0	0	0	0	1	1	1	1	1	1	1	1
Poland	0	0	0	0	0	1	1	1	1	1	1	1
Portugal	0	0	0	0	1	1	1	1	1	1	1	1
Romania	0	0	0	0	0	0	0	0	1	1	1	1
Slovak Republic	0	0	0	0	0	1	1	1	1	1	1	1
Slovenia	0	0	0	0	0	1	1	1	1	1	1	1
Spain	0	0	0	0	1	1	1	1	1	1	1	1
Sweden	0	0	0	0	1	1	1	1	1	1	1	1
Switzerland	0	0	0	0	0	0	0	1	1	1	1	1
Türkiye	0	0	0	1	0	0	1	1	1	1	1	1
United Kingdom	1	1	1	1	1	1	1	1	1	1	1	1
United States	1	1	1	1	1	1	1	1	1	1	1	1

	TIME_PERIOD										
Reference.area	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Australia	0	0	1	0	0	1	0	0	0	0	0
Austria	1	1	1	1	1	1	1	1	1	1	1
Belgium	1	1	1	1	1	1	1	1	1	1	1
Bulgaria	1	1	1	1	1	1	1	1	1	1	1
Canada	1	1	1	1	1	1	1	1	1	1	0
Chile	0	1	0	1	0	1	0	0	1	1	0
Colombia	0	0	0	0	0	0	1	1	0	0	0
Costa Rica	0	0	0	0	0	0	1	0	0	0	0
Croatia	1	1	1	1	1	1	1	1	1	1	1
Czechia	1	1	1	1	1	1	1	1	1	1	1
Denmark	1	1	1	1	1	1	1	1	1	1	1
Estonia	1	1	1	1	1	1	1	1	1	1	1
Finland	1	1	1	1	1	1	1	1	1	1	1
France	1	1	1	1	1	1	1	1	1	1	1
Germany	1	1	1	1	1	1	1	1	1	1	1
Greece	1	1	1	1	1	1	1	1	1	1	1
Hungary	1	1	1	1	1	1	1	1	1	1	1
Iceland	1	1	1	1	1	1	1	0	0	0	0
Ireland	1	1	1	1	1	1	1	1	1	1	1
Israel	1	1	1	1	1	1	1	1	1	1	0
Italy	1	1	1	1	1	1	1	1	0	1	1
Japan	0	1	0	0	1	0	0	1	0	0	0
Korea	1	0	1	0	1	0	1	0	1	0	1
Latvia	1	1	1	1	1	1	1	1	1	1	1
Lithuania	1	1	1	1	1	1	1	1	1	1	1
Luxembourg	1	1	1	1	1	1	1	1	1	1	1
Mexico	0	0	0	0	0	0	0	0	0	0	0
Netherlands	1	1	1	1	1	1	1	1	1	1	1
New Zealand	1	1	1	1	1	1	1	1	1	1	1
Norway	1	1	1	1	1	1	1	1	1	0	0
Poland	1	1	1	1	1	1	1	1	1	1	1
Portugal	1	1	1	1	1	1	1	1	1	1	1
Romania	1	1	1	1	1	1	1	1	1	1	1
Slovak Republic	1	1	1	1	1	1	1	1	1	1	1
Slovenia	1	1	1	1	1	1	1	1	1	1	1

```
Spain               1    1    1    1    1    1    1    1    1    1    1
Sweden              1    1    1    1    1    1    1    1    1    1    1
Switzerland         1    1    1    1    1    1    1    1    1    1    0
Türkiye             1    1    1    1    1    1    1    1    1    1    0
United Kingdom      1    1    1    1    1    1    1    1    0    0    0
United States       1    1    1    1    1    1    1    1    1    1    0
```

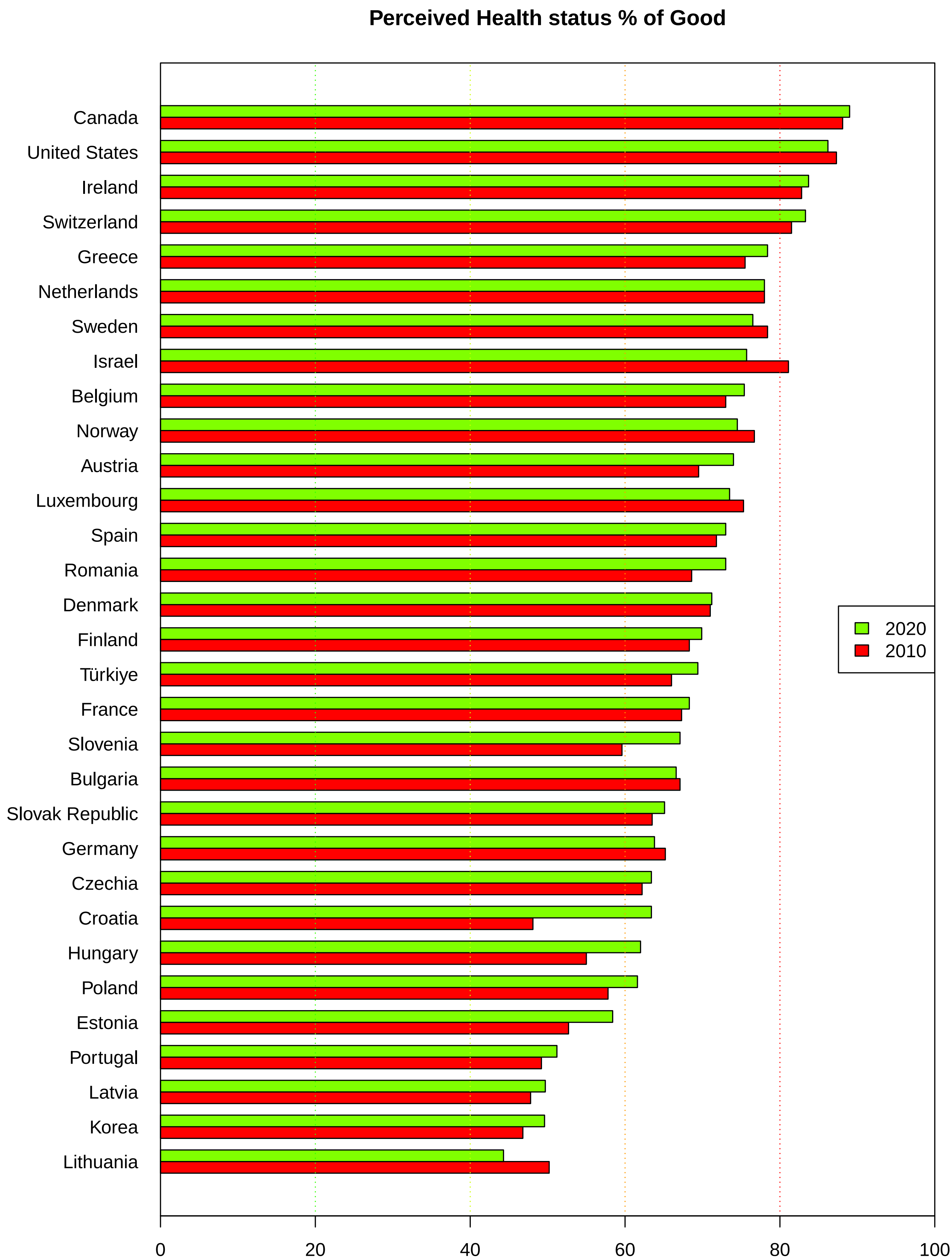
```r
## 국가별로 연도가 빠진 것이 있는지 check
timetable <- as.data.frame(table(status_Phs[status_Phs$TIME_PERIOD%in% c(2010,2020),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

```
   TIME_PERIOD Reference.area Freq
9         2010          Chile    0
30        2020        Iceland    0
36        2020          Italy    0
38        2020          Japan    0
49        2010    New Zealand    0
72        2020 United Kingdom    0
```

```r
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Phs[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Phs
status_Phs[status_Phs$TIME_PERIOD%in% c(2010,2020),] -> status_Phs
status_Phs[!status_Phs$Reference.area%in%excluded,]-> status_Phs
status_Phs %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Phs
mat_Phs <- mat_Phs[order(mat_Phs$`2020`),] ### 2020년도 good 비율 높은 순서대로 정렬
mat_Phs %>% str
```

```
'data.frame':   31 obs. of  3 variables:
 $ Reference.area: chr  "Lithuania" "Korea" "Latvia" "Portugal" ...
 $ 2010          : num  50.2 46.8 47.8 49.2 52.7 57.8 55 48.1 62.2 65.2 ...
 $ 2020          : num  44.3 49.6 49.7 51.2 58.4 61.6 62 63.4 63.4 63.8 ...
```

```r
mat_Phs[,2:3]%>% as.matrix %>% t -> matval
colnames(matval) <- mat_Phs$Reference.area
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(4)[1:2],horiz = T, las=1,
                   xlim=c(0,100),
                   main = "Perceived Health status % of Good",
                   args.legend = list(x="right"),
                   legend.text = c(2010,2020) )+
abline(v=c(20,40,60,80), lty=3,col=rainbow(10)[4:1])+
box()
```



Perceived Health status % of Good
Canada
United States
Ireland
Switzerland
Greece
Netherlands
Sweden
Israel
Belgium
Norway
Austria
Luxembourg
Spain
Romania
Denmark
Finland
Türkiye
France
Slovenia
Bulgaria
Slovak Republic
Germany
Czechia
Croatia
Hungary
Poland
Estonia
Portugal
Latvia
Korea
Lithuania
2020
2010
0
20
40
60
80
100

```
numeric(0)
```

1.2.5 Abscence from work due to illness

Absence from work due to illness항목은 두가지가 있다 이 중에서 Self-reported absence from work due to illness 항목에 대해서 조사해보자

```
status_Ab <- status[status$Measure=="Self-reported absence from work due to illness"
                    ,]
checkvar(status_Ab)
```

```
5 ) 27   REF_AREA
6 ) 27   Reference.area
31 ) 53   TIME_PERIOD
33 ) 96   OBS_VALUE
37 ) 4   OBS_STATUS
38 ) 4   Observation.status
39 ) 2   OBS_STATUS2
40 ) 2   Observation.status.2
```

```
checkvar2(status_Ab)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : SAWI
10 ) Measure : Self-reported absence from work due to illness
11 ) UNIT_MEASURE : D_Y_PS
12 ) Unit.of.measure : Days per year per person
13 ) AGE : _T
14 ) Age : Total
15 ) SEX : _T
16 ) Sex : Total
```

```
## status_Ab$Observation.status %>% unique
## status_Ab[,c("Reference.area","Observation.status")] %>% table
```

국가별/연도별 측정치가 한개씩인지 확인 (다른교란변수가 없는지) 그리고 연도별 측정 횟수가 가장 많은 연도를 알아보자 -> 2014,2019 선택

```
status_Ab[,c("Reference.area","TIME_PERIOD")] %>% table %>%
  as.data.frame() -> tempdf
tempdf$Freq %>% unique  # all values are 0 or 1
```

```
[1] 0 1
```

```r
tempdf[tempdf$Freq==1,"TIME_PERIOD"]   %>% table
```

```
.
1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985
   1    1    1    1    1    1    2    2    3    2    2    2    2    3    2    2
1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001
   2    3    3    4    5    3    6    4    5    6    5    5    5    5    7    5
2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017
   6    5    6    8    6    8   10   10    9    7   10    7   21    7   10    8
2018 2019 2020 2021 2022
  10   18   10    6    3
```

```r
timetable <- as.data.frame(table(status_Ab[status_Ab$TIME_PERIOD%in% c(2014,2019),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

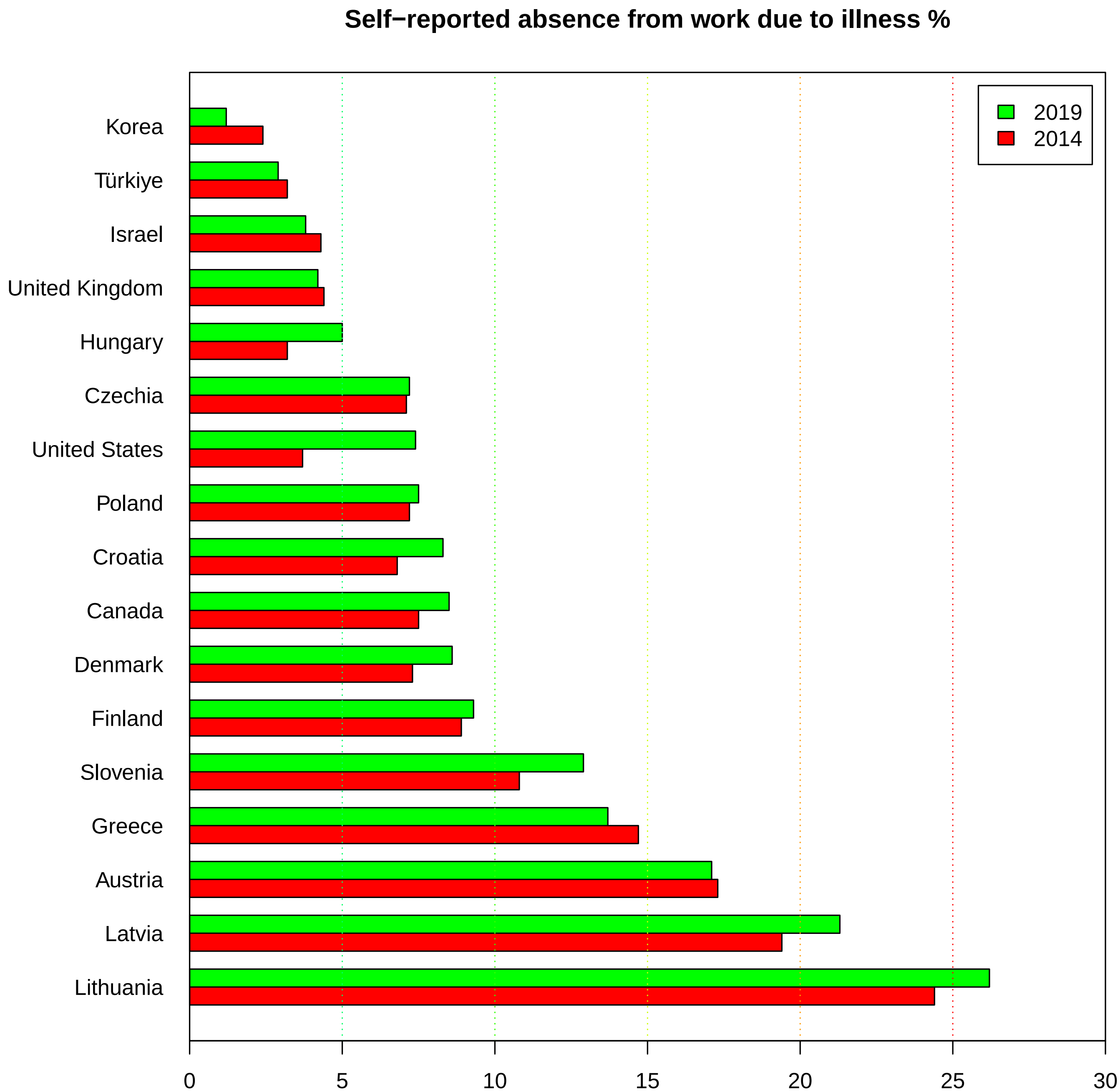
```
   TIME_PERIOD Reference.area Freq
2         2019      Australia    0
7         2014     Costa Rica    0
16        2019        Estonia    0
20        2019         France    0
38        2019          Spain    0
```

```r
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Ab[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Ab
status_Ab[status_Ab$TIME_PERIOD%in% c(2014,2019),] -> status_Ab
status_Ab[!status_Ab$Reference.area%in%excluded,]-> status_Ab
status_Ab %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Ab
mat_Ab <- mat_Ab[order(-mat_Ab$`2019`),] ### 2019년도 낮은 순서대로 정렬
mat_Ab %>% str
```

```
'data.frame':   17 obs. of  3 variables:
 $ Reference.area: chr  "Lithuania" "Latvia" "Austria" "Greece" ...
 $ 2014          : num  24.4 19.4 17.3 14.7 10.8 8.9 7.3 7.5 6.8 7.2 ...
 $ 2019          : num  26.2 21.3 17.1 13.7 12.9 9.3 8.6 8.5 8.3 7.5 ...
```

```r
mat_Ab[,2:3]%>% as.matrix %>% t -> matval
colnames(matval) <- mat_Ab$Reference.area
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(3)[1:2],horiz = T, las=1,
                   xlim=c(0,30),
                   main = "Self-reported absence from work due to illness %",
#                   args.legend = list(x="topright"),
```

```
                    legend.text = c(2014,2019))
abline(v=c(5,10,15,20,25), lty=3,col=rainbow(10)[5:1])
box()
```



1.2.6 Potential years of life lost

Potential years of life lost 항목에 대해서 변수를 조사해 보자

```r
status_Pl <- status[status$Measure=="Potential years of life lost" &
                      status$Sex=="Total",]
checkvar(status_Pl)
```

```
5 ) 46   REF_AREA
6 ) 46   Reference.area
19 ) 49   DEATH_CAUSE
20 ) 49   Cause.of.death
31 ) 62   TIME_PERIOD
33 ) 20008   OBS_VALUE
37 ) 3   OBS_STATUS
38 ) 3   Observation.status
```

```r
checkvar2(status_Pl)
```

```
1 ) STRUCTURE : DATAFLOW
2 ) STRUCTURE_ID : OECD.ELS.HD:DSD_HEALTH_STAT@DF_HEALTH_STATUS(1.0)
3 ) STRUCTURE_NAME : Health status
4 ) ACTION : I
7 ) FREQ : A
8 ) Frequency.of.observation : Annual
9 ) MEASURE : PYLL
10 ) Measure : Potential years of life lost
11 ) UNIT_MEASURE : Y_10P5PS
12 ) Unit.of.measure : Years per 100 000 persons
13 ) AGE : Y75
14 ) Age : 75 years
15 ) SEX : _T
16 ) Sex : Total
39 ) OBS_STATUS2 :
40 ) Observation.status.2 :
```

```r
status_Pl$Cause.of.death %>% unique
```

```
 [1] "External causes of mortality"
 [2] "Malignant neoplasms of colon, rectum and anus"
 [3] "Leukemia"
 [4] "Diseases of the genitourinary system"
 [5] "Transport Accidents"
 [6] "Accidental falls"
 [7] "Chronic liver diseases and cirrhosis"
 [8] "Certain conditions originating in the perinatal period"
 [9] "Diseases of the respiratory system"
[10] "Malignant melanoma of skin"
[11] "Endocrine, nutritional and metabolic diseases"
[12] "Diseases of the nervous system"
[13] "Diabetes mellitus"
[14] "Neoplasms"
```

```
[15] "Alzheimer's disease"
[16] "Malignant neoplasms of bladder"
[17] "Tuberculosis"
[18] "Accidental poisoning"
[19] "Accidents"
[20] "Dementia"
[21] "Ischaemic heart diseases"
[22] "Chronic obstructive pulmonary diseases"
[23] "Malignant neoplasms"
[24] "Assault"
[25] "HIV-AIDS"
[26] "Total"
[27] "Symptoms, signs, ill-defined causes"
[28] "Malignant neoplasms of pancreas"
[29] "Asthma"
[30] "Peptic ulcer"
[31] "Mental and behavioural disorders"
[32] "Malignant neoplasms of stomach"
[33] "Malignant neoplasms of trachea, bronchus, lung"
[34] "Diseases of the musculoskeletal system and connective tissue"
[35] "Certain infectious and parasitic diseases"
[36] "Diseases of the digestive system"
[37] "Congenital malformations, deformations and chromosomal abnormalities"
[38] "Diseases of the circulatory system"
[39] "Codes for special purposes: COVID-19"
[40] "Acute myocardial infarction"
[41] "Diseases of the skin and subcutaneous tissue"
[42] "Hodgkin's disease"
[43] "Cerebrovascular diseases"
[44] "Pneumonia"
[45] "Parkinson's disease"
[46] "Malignant neoplasms of liver"
[47] "Influenza"
[48] "Diseases of the blood and blood-forming organs"
[49] "Intentional self-harm"
```

```
status_Pl <- status[status$Measure=="Potential years of life lost" &
                      status$Sex=="Total" &
                      status$Cause.of.death=="Total",]
checkvar(status_Pl)
```

```
5 ) 46   REF_AREA
6 ) 46   Reference.area
31 ) 62   TIME_PERIOD
33 ) 2280   OBS_VALUE
37 ) 3   OBS_STATUS
38 ) 3   Observation.status
```

Sex, Cause of death는 전부 Total로 하자. 국가별 사망원인별 통계도 필요하면 할 수 있다.

```r
status_Pl[,c("Reference.area","TIME_PERIOD")] %>% table %>%
  as.data.frame() -> tempdf
tempdf$Freq %>% unique  # all values are 0 or 1
```

```
[1] 0 1
```

```r
tempdf[tempdf$Freq==1,"TIME_PERIOD"]   %>% table
```

```
.
1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975
  24   26   26   27   27   27   28   29   29   31   29   29   30   29   29   30
1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991
  30   31   30   31   34   36   37   34   34   40   41   40   40   40   43   43
1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007
  44   43   44   44   45   44   44   45   44   45   45   45   44   43   44   45
2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021
  45   46   46   45   46   46   46   44   44   41   39   38   31    9
```

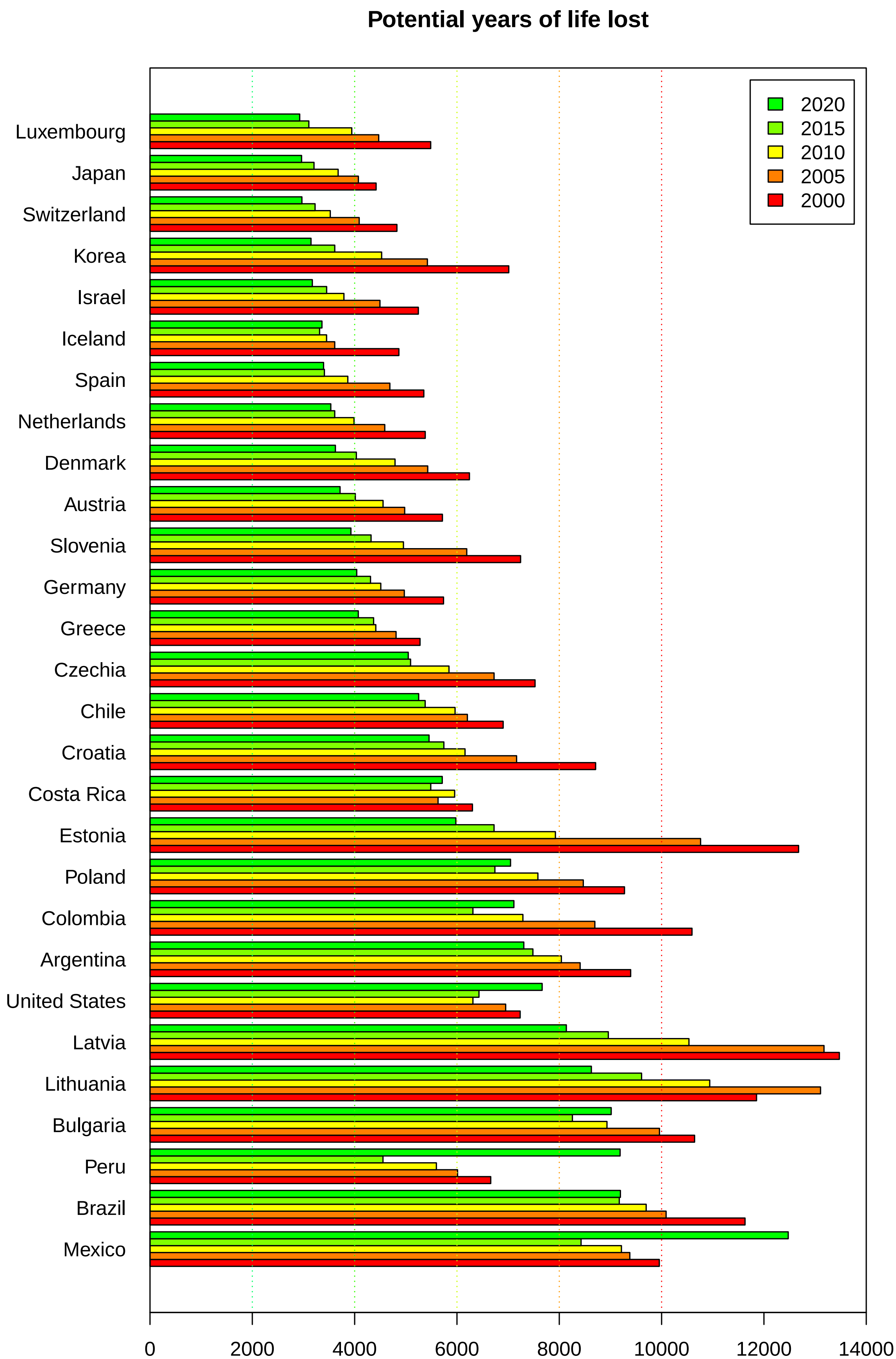
측정연도는 2000,2005,2010,2015,2020 로 하도록 하자

```r
##status_Am[,c("Reference.area","TIME_PERIOD")] %>% table
## 국가별로 연도가 빠진 것이 있는지 check
timetable <- as.data.frame(table(status_Pl[status_Pl$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),
              c("TIME_PERIOD","Reference.area")]))
timetable[timetable$Freq==0,]
```

```
    TIME_PERIOD  Reference.area Freq
7          2005       Australia    0
20         2020         Belgium    0
35         2020          Canada    0
74         2015         Finland    0
80         2020          France    0
95         2020         Hungary    0
105        2020         Ireland    0
115        2020           Italy    0
155        2020     New Zealand    0
160        2020          Norway    0
172        2005        Portugal    0
175        2020        Portugal    0
180        2020         Romania    0
185        2020          Russia    0
189        2015 Slovak Republic    0
190        2020 Slovak Republic    0
200        2020    South Africa    0
210        2020          Sweden    0
216        2000         Türkiye    0
217        2005         Türkiye    0
```

```
220        2020         Türkiye    0
221        2000  United Kingdom    0
```

```
timetable[timetable$Freq==0,"Reference.area"] -> excluded
status_Pl[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> status_Pl
status_Pl[status_Pl$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),] -> status_Pl
status_Pl[!status_Pl$Reference.area%in%excluded,]-> status_Pl
status_Pl %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_Pl
mat_Pl <- mat_Pl[order(-mat_Pl$`2020`),] ### 2020년도 낮은 순서대로 정렬
mat_Pl[,2:6]%>% as.matrix %>% t -> matval
colnames(matval) <- mat_Pl$Reference.area
par(mar=c(4,8,4,2)+0.1)
matval %>% barplot(,beside = T,col=rainbow(12)[1:5],horiz = T, las=1,
                   xlim=c(0,14000),
                   main = "Potential years of life lost",
#                   args.legend = list(x="topright"),
                   legend.text = c(2000,2005,2010,2015,2020) )
abline(v=c(2000,4000,6000,8000,10000), lty=3,col=rainbow(10)[5:1])
box()
```


Potential years of life lost
2020
2015
2010
2005
2000
Luxembourg
Japan
Switzerland
Korea
Israel
Iceland
Spain
Netherlands
Denmark
Austria
Slovenia
Germany
Greece
Czechia
Chile
Croatia
Costa Rica
Estonia
Poland
Colombia
Argentina
United States
Latvia
Lithuania
Bulgaria
Peru
Brazil
Mexico
0
2000
4000
6000
8000
10000
12000
14000

Potential years of life lost 계산법

totalling deaths occurring at each age and multiplying this with the number of remaining years to live up to a selected age limit (75 years old is used in OECD Health Statistics).

통계적 의미 = 젊은 나이에 죽는 사람이 많을 수록 수치가 올라간다.

1.2.7 데이터 요약 매트릭스 만드는 함수와 그래프 그리는 함수

측정변수가 Potential years of life lost 인 경우에 한하여 사망원인이 다양하기 때문에 사망원인에 따른 그래프를 별도로 그리고 싶다. 아예 사망원인을 변수로 넣으면 자동으로 value matrix 를 만드는 함수와 매트릭스를 넣으면 그래프를 그려주는 함수를 만들어보자

```
PYLLmat <- function(COD="Total", Sex="Total"){
 PL <- status[status$Measure=="Potential years of life lost" &
                     status$Sex==Sex &
                     status$Cause.of.death==COD,]
 TT <- as.data.frame(table(PL[PL$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),
             c("TIME_PERIOD","Reference.area")]))
 TT[TT$Freq==0,"Reference.area"] -> excluded
 PL[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> PL
 PL[PL$TIME_PERIOD%in% c(2000,2005,2010,2015,2020),] -> PL
 PL[!PL$Reference.area%in%excluded,]-> PL
 PL %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_PL
 mat_PL <- mat_PL[order(-mat_PL$`2020`),] ### 2020년도 낮은 순서대로 정렬
 mat_PL[,-1]%>% as.matrix %>% t -> mat
 colnames(mat) <- mat_PL$Reference.area
 return(mat)
}
PYLLmat(COD="Transport Accidents") %>% t
```

```
              2000  2005  2010  2015  2020
Brazil        588.8 641.7 716.5 609.1 488.6
United States 538.9 524.7 376.7 380.3 416.5
Colombia      601.7 484.3 406.2 457.5 355.1
Mexico        524.3 540.8 487.0 428.4 343.2
Costa Rica    569.7 448.6 424.4 504.1 330.6
Poland        547.4 445.5 342.4 264.7 278.5
Chile         447.1 420.7 361.8 327.9 270.9
Latvia        946.5 673.7 383.2 345.6 250.9
Croatia       525.3 449.7 315.9 287.9 205.1
Greece        590.6 564.2 394.0 423.5 200.9
Lithuania     702.6 846.5 328.5 283.8 199.0
Bulgaria      355.1 335.8 297.1 287.7 198.0
Czechia       462.3 381.5 258.5 235.2 179.0
Argentina     372.2 324.5 383.8 371.7 175.3
Slovenia      484.5 440.6 218.6 199.8 153.4
Estonia       568.6 501.2 206.9 194.3 152.8
```

```
Peru          254.0 235.4 282.4 271.2 152.1
Korea         708.8 390.1 289.5 196.1 115.2
Israel        254.7 223.1 164.1 126.8 110.1
Austria       382.9 301.9 195.5 151.0 101.8
Spain         485.4 338.4 158.0 114.3  97.2
Germany       330.4 218.7 149.6 121.7  92.0
Luxembourg    431.0 274.0 149.6 114.1  90.4
Denmark       301.4 214.3 141.3 101.1  86.7
Iceland       423.4 208.3 134.6 156.8  82.8
Netherlands   228.8 143.3 113.8  86.3  80.3
Switzerland   253.2 159.5 117.6  84.8  60.8
Japan         235.1 168.6 107.8  79.2  52.5
```

```
PYLLmat(COD="Codes for special purposes: COVID-19") %>% t
```

```
                 [,1]
Peru           3956.8
Mexico         2562.8
Colombia       1010.2
Brazil          969.7
Argentina       909.8
Bulgaria        747.4
Chile           626.9
United States   523.9
Poland          431.9
Costa Rica      394.0
Croatia         325.9
United Kingdom  304.3
Lithuania       294.9
Spain           262.9
Czechia         211.4
Slovenia        199.3
Greece          163.1
Netherlands     153.5
Latvia          152.6
Israel          142.4
Luxembourg      122.5
Austria         106.7
Switzerland     103.0
Germany          67.4
Estonia          32.0
Denmark          31.4
Finland          20.0
Iceland           8.0
Japan             4.4
Australia         4.0
Korea             3.9
```

```r
PYLLmat <- function(COD="Total", Sex="Total", period=c(2000,2005,2010,2015,2020)){
 PL <- status[status$Measure=="Potential years of life lost" &
                     status$Sex==Sex &
                     status$Cause.of.death==COD,]
 TT <- as.data.frame(table(PL[, c("TIME_PERIOD","Reference.area")]))
 TT <- TT[TT$TIME_PERIOD%in%period,]
 TT[TT$Freq==0,"Reference.area"] -> excluded
 PL[,c("Reference.area","TIME_PERIOD","OBS_VALUE")]-> PL
 PL[PL$TIME_PERIOD%in% period,] -> PL
 PL[!PL$Reference.area%in%excluded,]-> PL
 PL %>% spread(key=TIME_PERIOD, value = OBS_VALUE) -> mat_PL
 lastvalue <- mat_PL[,length(mat_PL)]
 mat_PL <- mat_PL[rev(order(lastvalue)),] ### 낮은 순서대로 정렬
 mat_PL[,-1] %>% as.matrix %>% t -> mat
 colnames(mat) <- mat_PL$Reference.area
 return(mat)
}
PYLLmat(COD="Total") %>% t
```

```
                 2000    2005    2010   2015    2020
Mexico         9955.0  9378.6  9214.5 8425.6 12474.8
Brazil        11630.4 10087.5  9698.0 9172.6  9194.6
Peru           6660.0  6011.9  5597.8 4554.0  9188.8
Bulgaria      10644.6  9958.1  8932.2 8256.5  9014.5
Lithuania     11855.1 13106.2 10940.5 9608.5  8626.9
Latvia        13474.0 13174.3 10534.1 8958.2  8137.5
United States  7235.3  6952.1  6312.1 6430.1  7664.8
Argentina      9394.9  8407.5  8040.6 7484.0  7306.2
Colombia      10593.8  8694.8  7288.6 6311.9  7112.5
Poland         9274.2  8469.0  7581.6 6741.4  7047.0
Estonia       12677.5 10761.7  7924.4 6725.1  5977.7
Costa Rica     6302.5  5630.4  5954.7 5487.9  5712.2
Croatia        8709.8  7165.3  6157.7 5743.9  5454.2
Chile          6902.7  6203.2  5962.9 5378.4  5251.5
Czechia        7526.3  6725.3  5844.2 5093.2  5047.6
Greece         5277.9  4809.6  4413.6 4370.8  4071.2
Germany        5737.3  4970.7  4510.9 4310.2  4040.6
Slovenia       7243.4  6191.1  4953.8 4320.8  3927.1
Austria        5715.7  4977.9  4555.5 4013.0  3715.0
Denmark        6243.3  5428.6  4789.4 4033.3  3623.7
Netherlands    5379.8  4588.7  3987.6 3610.6  3533.8
Spain          5352.0  4688.6  3866.3 3408.6  3392.6
Iceland        4864.8  3610.6  3451.5 3315.1  3361.1
Israel         5245.0  4494.8  3790.4 3452.8  3172.9
Korea          7012.7  5422.2  4527.7 3611.4  3146.9
```

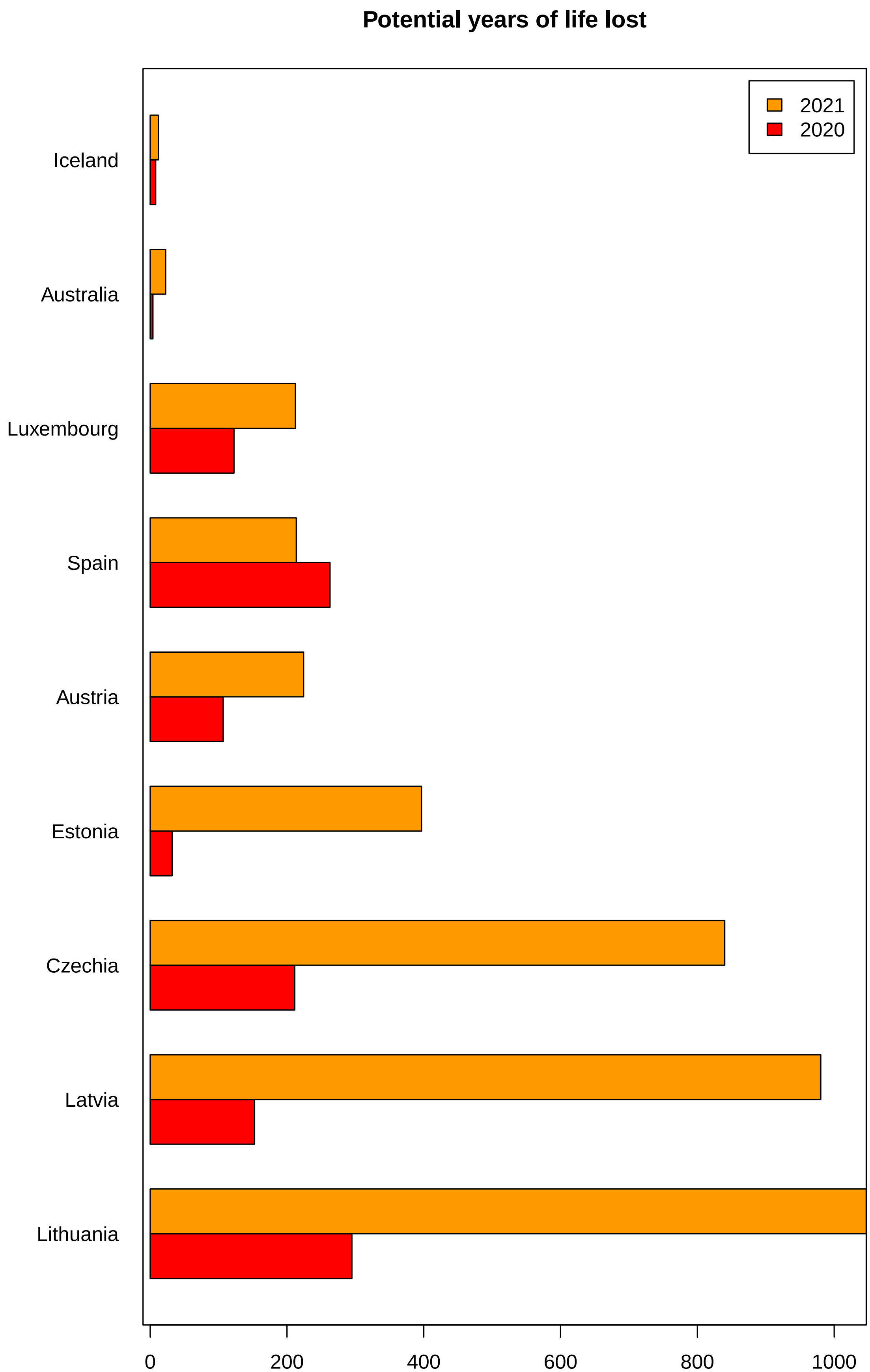
```
Switzerland    4826.2  4089.0  3523.6 3226.7  2969.2
Japan          4419.8  4072.8  3677.6 3205.6  2962.5
Luxembourg     5486.5  4470.5  3944.1 3105.0  2926.2
```

```r
PYLLmat(COD="Codes for special purposes: COVID-19", period=c(2020,2021)) %>% t
```

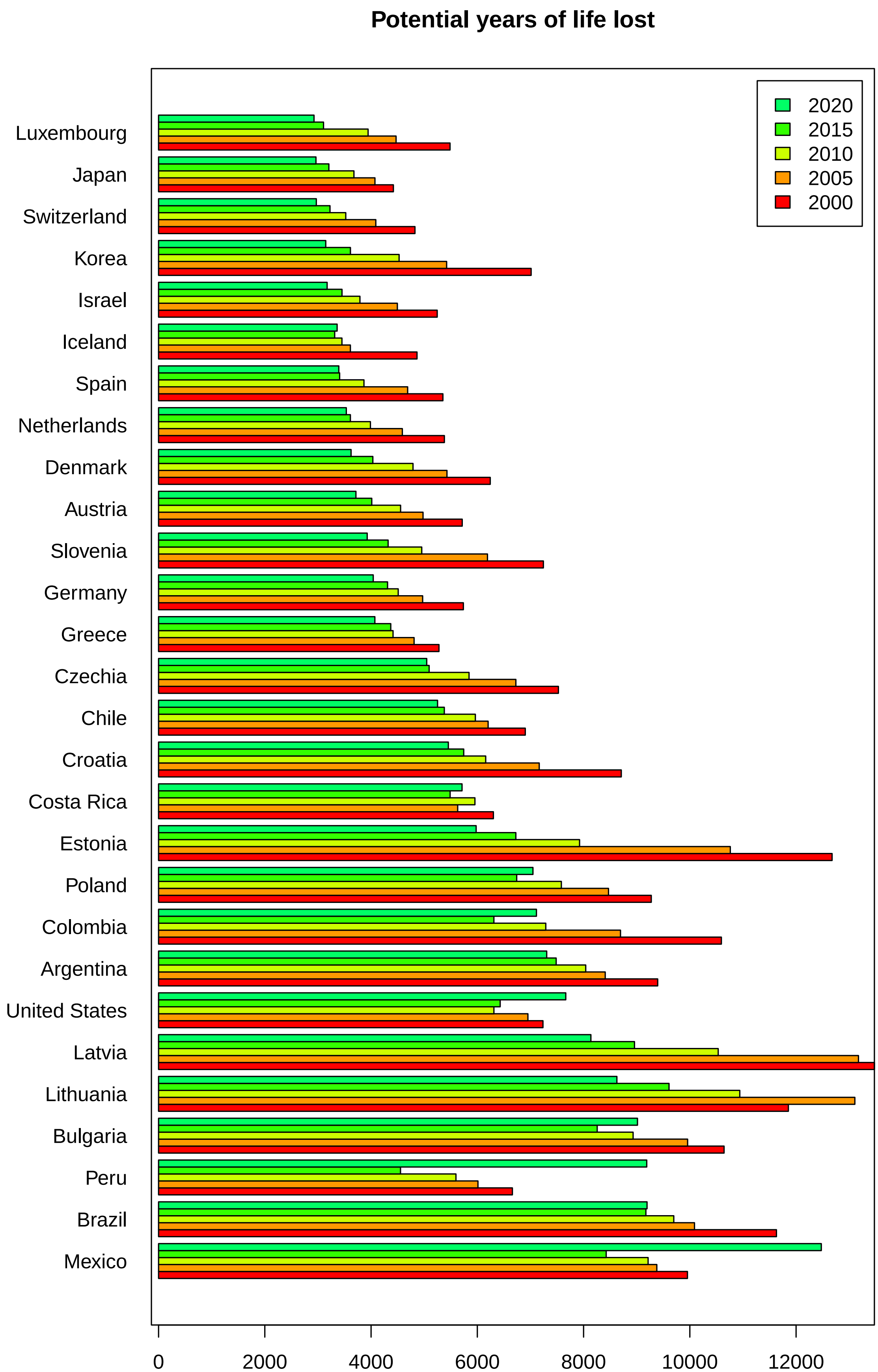
```
             2020   2021
Lithuania  294.9 1046.9
Latvia     152.6  980.4
Czechia    211.4  840.0
Estonia     32.0  396.7
Austria    106.7  224.3
Spain      262.9  213.6
Luxembourg 122.5  212.3
Australia    4.0   22.6
Iceland      8.0   12.0
```

Barplot을 만들거나 latex table을 만들 수 있다.

```r
oecdplot <- function(mat=matrix(1:4,nrow=2)){
  par(mar=c(4,8,4,2)+0.1)
  nbar <- nrow(mat)
  col.bar <- rainbow(10)[1:nbar]
  period <- as.integer(rownames(mat))
  mat %>% barplot(,beside = T,col=col.bar,horiz = T, las=1,
                  main = "Potential years of life lost",
#                  args.legend = list(x="topright"),
                  legend.text = period )
box()
}
PYLLmat(COD="Codes for special purposes: COVID-19",
        period=c(2020,2021)) %>% oecdplot
```


Potential years of life lost
2021
2020
Iceland
Australia
Luxembourg
Spain
Austria
Estonia
Czechia
Latvia
Lithuania
0
200
400
600
800
1000

```
PYLLmat(COD="Total") %>% oecdplot
```


Potential years of life lost
2020
2015
2010
2005
2000
Luxembourg
Japan
Switzerland
Korea
Israel
Iceland
Spain
Netherlands
Denmark
Austria
Slovenia
Germany
Greece
Czechia
Chile
Croatia
Costa Rica
Estonia
Poland
Colombia
Argentina
United States
Latvia
Lithuania
Bulgaria
Peru
Brazil
Mexico
0
2000
4000
6000
8000
10000
12000

```r
oecdltx <- function(mat=matrix(1:4,nrow=2)){
  mat %>% t %>% kableExtra::kable(,format = "latex") %>% cat
}
PYLLmat(COD="Total") %>% oecdltx
```

	2000	2005	2010	2015	2020
Mexico	9955.0	9378.6	9214.5	8425.6	12474.8
Brazil	11630.4	10087.5	9698.0	9172.6	9194.6
Peru	6660.0	6011.9	5597.8	4554.0	9188.8
Bulgaria	10644.6	9958.1	8932.2	8256.5	9014.5
Lithuania	11855.1	13106.2	10940.5	9608.5	8626.9
Latvia	13474.0	13174.3	10534.1	8958.2	8137.5
United States	7235.3	6952.1	6312.1	6430.1	7664.8
Argentina	9394.9	8407.5	8040.6	7484.0	7306.2
Colombia	10593.8	8694.8	7288.6	6311.9	7112.5
Poland	9274.2	8469.0	7581.6	6741.4	7047.0
Estonia	12677.5	10761.7	7924.4	6725.1	5977.7
Costa Rica	6302.5	5630.4	5954.7	5487.9	5712.2
Croatia	8709.8	7165.3	6157.7	5743.9	5454.2
Chile	6902.7	6203.2	5962.9	5378.4	5251.5
Czechia	7526.3	6725.3	5844.2	5093.2	5047.6
Greece	5277.9	4809.6	4413.6	4370.8	4071.2
Germany	5737.3	4970.7	4510.9	4310.2	4040.6
Slovenia	7243.4	6191.1	4953.8	4320.8	3927.1
Austria	5715.7	4977.9	4555.5	4013.0	3715.0
Denmark	6243.3	5428.6	4789.4	4033.3	3623.7
Netherlands	5379.8	4588.7	3987.6	3610.6	3533.8
Spain	5352.0	4688.6	3866.3	3408.6	3392.6
Iceland	4864.8	3610.6	3451.5	3315.1	3361.1
Israel	5245.0	4494.8	3790.4	3452.8	3172.9
Korea	7012.7	5422.2	4527.7	3611.4	3146.9
Switzerland	4826.2	4089.0	3523.6	3226.7	2969.2
Japan	4419.8	4072.8	3677.6	3205.6	2962.5
Luxembourg	5486.5	4470.5	3944.1	3105.0	2926.2